

ANSWERS & EXPLANATIONS
GENERAL STUDIES (P) TEST – 6332

- Q 1. C** • Haemophilia is a genetic disorder (specifically, an X-linked recessive Mendelian disorder) that impairs the body's ability to control blood clotting. **Hence option 1 is correct.**
- Thalassemia is an inherited blood disorder (an autosomal recessive Mendelian disorder) characterized by abnormal or insufficient hemoglobin production. **Hence option 2 is correct.**
 - Sickle Cell Anaemia is also an inherited blood disorder (an autosomal recessive Mendelian disorder) caused by a mutation in the gene responsible for hemoglobin synthesis, leading to misshapen red blood cells. **Hence option 3 is correct.**
 - Arthritis is classified as an autoimmune disorder. Arthritis is the swelling and tenderness of one or more joints, with primary symptoms being joint pain, stiffness, and reduced range of motion, often worsening with age. The most common forms are osteoarthritis (wear-and-tear) and rheumatoid arthritis (autoimmune). Treatment includes medication, exercise, and sometimes surgery. **Hence option 4 is not correct.**

Q 2. C Thomson proposed that:

- An atom consists of a positively charged sphere and the electrons are embedded in it.
- **The negative and positive charges are equal in magnitude. So, the atom as a whole is electrically neutral. Hence statement 1 is correct.**
- Although Thomson's model explained that atoms are electrically neutral, the results of experiments carried out by other scientists could not be explained by this model.

On the basis of his experiment, Rutherford put forward the nuclear model of an atom, which had the following features:

- **There is a positively charged center in an atom called the nucleus. Nearly all the mass of an atom resides in the nucleus. Hence statement 2 is correct.**
- The electrons revolve around the nucleus in circular paths.
- The size of the nucleus is very small as compared to the size of the atom.

Drawbacks of Rutherford's model of the atom

- The revolution of the electron in a circular orbit is not expected to be stable. Any particle in a circular orbit would undergo acceleration. During acceleration, charged particles would radiate energy. Thus, the revolving electron would lose energy and finally fall into the nucleus. If this were so, the atom should be highly unstable, and hence matter would not exist in the form that we know. We know that atoms are quite stable.

In order to overcome the objections raised against Rutherford's model of the atom, **Neils Bohr put forward the following postulates about the model of an atom:**

- **Only certain special orbits known as discrete orbits of electrons are allowed inside the atom.**

- While revolving in discrete orbits the electrons do not radiate energy.

Q 3. A • Scientists have successfully grown a “human embryo” in the lab without using an egg or sperm. They used a mix of stem cells — early cells that have the ability to differentiate into other types of cells — that were able to spontaneously assemble into an embryo-like structure, mimicking molecular characteristics of an early embryo.

- **ASCs are undifferentiated cells found living within specific differentiated tissues in our bodies** that can renew themselves or generate new cells that can replenish dead or damaged tissue. The term “**somatic stem cell**” is used to refer to adult stem cells. The term “somatic” refers to non-reproductive cells in the body (eggs or sperm). ASCs are typically scarce in native tissues which has rendered them difficult to study and extract for research purposes.

Characteristic	Adult Stem Cells (ASCs)	Embryonic Stem Cells (ESCs)
Source	Found in various tissues and organs of adults	Derived from the inner cell mass of a blastocyst during in vitro fertilization
Pluripotency/Multipotency	Usually multipotent, with a limited differentiation potential	Pluripotent, capable of differentiating into almost any cell type in the body
Stage of Development	Present throughout one's lifetime	Present during the early stages of embryonic development
Origin	Naturally occurring in the body	Derived from embryos, not naturally occurring
Application in Therapy	Used in certain therapies, but differentiation potential is more limited	Considered highly versatile and has the potential for a broader range of therapeutic applications

- **Embryonic Stem Cells (ESCs) are formed during days 3-5 following fertilization and before implantation, the embryo** (at this stage, called a blastocyst), contains an inner cell mass that is capable of generating all the specialized tissues that make up the human body.
- **Embryonic Stem Cells (ESCs) are derived from the inner cell mass of a blastocyst** following in vitro fertilization, whereas **Adult Stem Cells (ASCs) are found in various tissues and organs. Hence statement 1 is correct.**
- **ESCs are pluripotent**, meaning they can differentiate into nearly any cell type, whereas **ASCs are often multipotent**, having a more limited differentiation potential. **Hence statement 2 is not correct.**
- **ASCs are found during the early stages of embryonic development**, whereas **ESCs Present throughout one's lifetime**. This statement is reversed. ASCs are found throughout one's lifetime in various tissues, while ESCs are present during the early stages of embryonic development. **Hence statement 3 is not correct.**
- **Hence option (a) is the correct answer.**

Q 4. C • **Myopia or shortsightedness** is an eye defect in which the person is unable to see far objects clearly. It occurs either due to the shifting of the retina or due to a decrease in the focal length of the lens. **It can be corrected by using a concave lens.**

- **Hypermetropia or long-sightedness** is an eye defect where affected person is not able to see close objects clearly. It occurs either due to shifting of the retina in a forward direction or when the focal length of lens increases. **It is corrected by using a convex lens.**
- **Presbyopia:** It is age-related eye defects where a person is not able to see both far-off and close objects clearly, it occurs due to the weakening of ciliary muscles and hardening of the eye lens. **It is corrected by using a bifocal lens which is a combination of both concave and convex lens.**

Q 5. B • Option (b) is correct.

- Work (W) is defined as the product of the force (F) and the displacement (s) in the direction of force.
- $W = F \times s$
- If force is 1 Newton (N), displacement is 1 metre (m) then work done is 1 Joule (J).
- Thus, work done by a force acting on an object is equal to the magnitude of the force multiplied by the distance moved in the direction of the force. Work has only magnitude and no direction.
- If there is no displacement, there is no work done even if the force is large. Thus, when the man pushes hard against a rigid brick wall, the force he exerts on the wall does no work, even though his muscles are alternatively contracting and relaxing and internal energy is being used up.

Q 6. D • Mycelia are the branching filaments of fungi that make up their vegetative parts. Researchers have them harnessed to create **biodegradable, fire-resistant, and insulative bricks**. They combine **husk, sawdust, and fungal spores** to create a fibrous network that solidifies into a **lightweight material**. Mycelium bricks can potentially **reduce the building sector's carbon footprint**, and they are relatively lightweight and **good insulators of heat**. Aside from potential use as **paneling material in interior designs**, researchers believe mycelium-based components can be used in **liquid filters, sports equipment, and printed circuit boards**.

- **Hence option (d) is the correct answer.**

Q 7. A Saturated vs Unsaturated Fatty Acids

- **Saturated fatty acids:** only single C-C bonds; pack tightly and are typically solid at room temperature (e.g., butter, ghee).
- **Unsaturated fatty acids:** one or more C=C double bonds; kinks reduce packing and are typically liquid at room temperature (e.g., olive oil, fish oil).

Analysis of Statements:

- **Statement (1):** Correct - Presence of double bonds defines unsaturation (mono- or polyunsaturated).
- **Statement (2):** Correct - By definition, saturated chains contain no double bonds.
- **Statement (3):** Correct - Double bonds introduce kinks, reducing intermolecular interactions and lowering melting points.
- **Statement (4):** Correct - Tight packing due to straight chains raises melting points, making them solid at room temperature.

Hence, the correct answer is **(a) 1, 2, 3 and 4.**

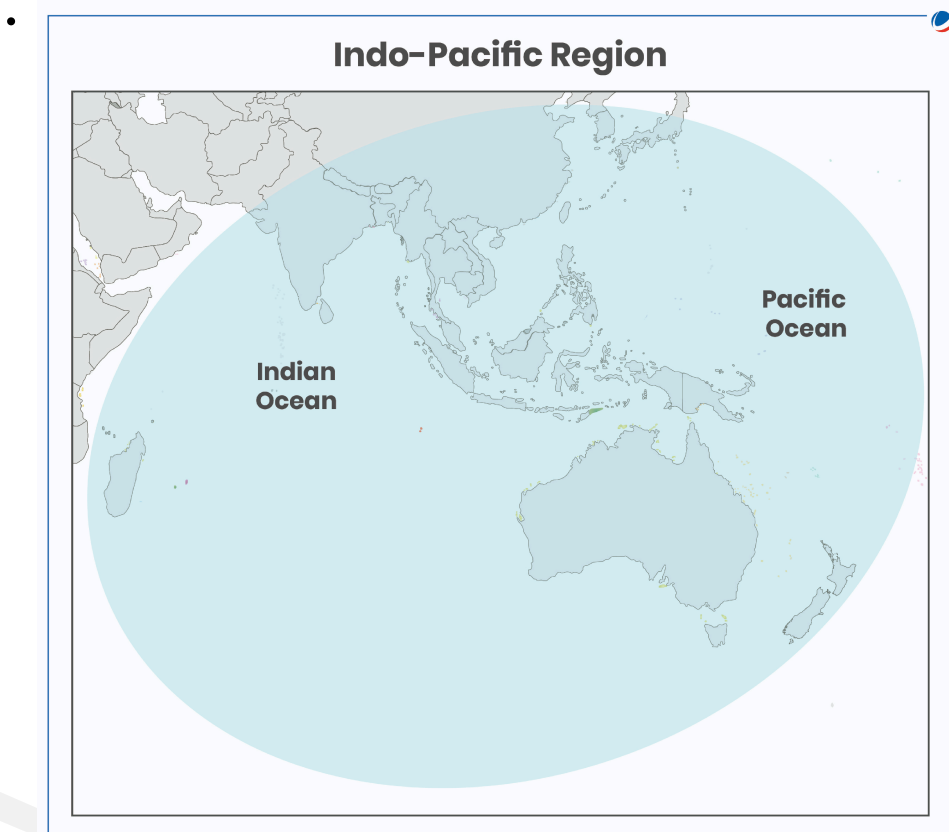
- Q 8. A** • **Next Generation Launch Vehicle (NGLV)** is a significant step towards the Government's vision of establishing & operating the Bharatiya Antariksh Station and towards developing capability for Indian Crewed Landing on the Moon by 2040.
- Next Generation Launch Vehicle (NGLV) **is designed to have a maximum payload capability of 30 tonnes to Low Earth Orbit, which also has a reusable first stage, hence it is partially reusable vehicle.** This new launch vehicle will feature reusability and modular green propulsion systems, promising reduced launch costs and improved access to space. **Hence, statement 1 is correct and 2 is not correct.**
 - Currently, India has achieved self-reliance in space transportation systems to launch satellites up to 10 tonne to Low Earth Orbit (LEO) and 4 tonne to Geo-Synchronous Transfer Orbit (GTO) through the currently operational PSLV, GSLV, LVM3 & SSLV launch vehicles.
 - The development of NGLV will enable national & commercial missions including launch of human spaceflight missions to Bharatiya Antariksh Station, Lunar/inter-planetary exploration missions along with communication & earth observation satellite constellations to Low Earth Orbit that will benefit the entire space ecosystem in the country.
- Q 9. C** • Osmium, atomic number **76**, is a **shiny, silver metal** that resists corrosion. It is the **densest of all** the elements and is twice as dense as lead. Its density of **22.6 g/cm³**. Osmium has only a few uses. **Hence, statement 1 is correct.**
- Bromine is a fairly abundant element and it is the **only non-metal** to exist in liquid form at room temperature. Mercury is the only metal to exist in a liquid state under standard temperature and pressure conditions. **Hence, statement 2 is correct.**
- Man-made elements** are generally radioactive elements that were not discovered occurring in nature but as artificially produced isotopes.
- **Technetium** was discovered by Emilio Segrè in 1937 in Italy. He investigated molybdenum from California which had been exposed to high energy radiation and he found technetium to be present and separated it. Today, this element is extracted from spent nuclear fuel rods in tonne quantities.
 - In 1937, it became the first element to be discovered by the synthesis in a laboratory. **Hence, statement 3 is correct.**
- Rare earth elements** are a group of **seventeen chemical elements** that occur together in the periodic table.
- The group consists of **yttrium and the 15 lanthanide elements** (lanthanum, cerium, praseodymium, neodymium, promethium, samarium, europium, gadolinium, terbium, dysprosium, holmium, erbium, thulium, ytterbium, and lutetium). **Hence, statement 4 is not correct.**
 - **Scandium is found in most rare earth element deposits and is sometimes classified as a rare earth element.**
 - These metals have many similar properties, and that often causes them to be found together in geologic deposits.
 - While Rare Earth elements are used in building consumer electronics, healthcare and transportation, they are especially important for governments because of their use in manufacturing defence equipment. At present, China refines approximately 80%-90% of the world's Rare Earths, thereby having substantial control over their supply.

- Q 10. B** • Plasma is the straw-colored, liquid component of blood that constitutes approximately 55% of the total blood volume. It is the largest part of blood, with the remaining 45% consisting of formed elements like red blood cells, white blood cells, and platelets. Plasma consists of about 90–92% water, 6–8% proteins (albumins, globulins, fibrinogen), and 1% other solutes like electrolytes and nutrients. **Hence, statement 1 is correct.**
- Red blood cells (RBCs) contain haemoglobin, an iron-rich protein that enables the transport of oxygen from the lungs to body tissues and returns carbon dioxide. Each haemoglobin molecule binds up to four oxygen molecules, making it highly efficient. It also gives blood its red color. **Hence, statement 2 is correct.**
 - White blood cells (WBCs) are primarily involved in the immune response, helping the body fight infections and diseases. The platelets (along with clotting factors found in the plasma) are the components responsible for the clotting of blood to prevent excessive bleeding. **Hence, statement 3 is not correct.**
- Q 11. B** • Carbon has several allotropes or different forms in which it can exist. These allotropes include **graphite, diamond, graphene, and fullerenes** which have very different properties. **Hence all the options are correct.**
- **Diamond** is a well-known allotrope of carbon that exhibits hardness and high dispersion of light. It is the hardest known natural mineral and finds applications in cutting, drilling, and jewelry, and as a potential semiconductor material.
 - **Graphite is an opaque**, a very good lubricant, a good conductor of electricity, and a thermal insulator.
 - **Graphene** is a single layer of carbon atoms arranged in one plane; layers of graphene make up graphite. Graphene is a material of interest due to its high electron mobility and its possible applications in electronics.
 - **Fullerenes** are a class of carbon allotropes in which carbon takes the form of a hollow sphere, ellipsoid, or tube. This class of materials includes carbon nanotubes, buckyballs, and the newly discovered nanobuds.
 - Other allotropes of Carbon include amorphous carbon, glassy carbon, carbon nanofoam, nanotubes, etc.
- Q 12. D** • Acids are a substance with particular chemical properties including **turning litmus red**, neutralizing alkalis, and dissolving some metals. An acid can **donate an H⁺ ion** and can remain energetically favourable after a loss of H⁺. They have a pH value of **less than 7**.
- Bases is a substance able to accept a hydrogen ion from an acid. They can **donate hydroxide ions (OH⁻)** in solution. They turn red litmus blue and have a **pH value greater than 7**.
- A special type of substances is used to test whether a substance is acidic or basic. These substances are known as indicators.
- **Litmus** is a natural indicator, **turmeric** is another such indicator.
 - There are many other natural materials like **red cabbage leaves**, coloured petals of some flowers such as Hydrangea, Petunia and Geranium, which indicate the presence of acid or base in a solution. These are called acid-base indicators or sometimes simply indicators.
 - **Olfactory indicators** are substances that change their smell when mixed with an acid or base, for example, **onion**, vanilla and clove.

- Hence (d) is the correct answer.

Q 13. B • **Recent Context:** Spain Joins the Indo-Pacific Oceans Initiative (IPOI)

- **About Indo-Pacific Oceans Initiative (IPOI)**
- **Genesis:** IPOI was launched by India at the East Asia Summit in 2019, held in Bangkok (Thailand). Hence, statement 1 is correct.
- **Background:** It builds upon the “Security and Growth for All in the Region” (SAGAR) initiative announced by India in 2015. Hence, statement 2 is correct.
- **IPOI is a non-treaty-based voluntary arrangement that promotes cooperation for a free and open Indo-Pacific and the rules-based regional order. Hence, statement 3 is not correct.**
- SAGAR encourages countries to cooperate and synergise efforts towards a safe, secure, and stable maritime domain.
- 7 pillars of IPO: It includes Maritime Security; Maritime Ecology, Capacity Building and Resource Sharing; Disaster Risk Reduction and Management, etc.



Q 14. C • **Recent Context:** RBI issued Reserve Bank of India (Priority Sector Lending – Targets and Classification) Directions, 2025.

- PSL ensure that vulnerable sections of society and underdeveloped areas get access to credit.
- Key Features of the Latest RBI PSL Guidelines**
- **Enhanced Compliance & External Audit:** RBI requires banks to obtain certification from external auditors (or CAG-empanelled auditors for specific entities like NCDC).
 - This audit ensures that the same underlying loan exposure is not claimed as PSL by both the originating bank and the intermediary (such as an NBFC or Cooperative).

- **Revised Sector Targets:** The PSL target for Small Finance Banks (SFBs) has been adjusted from 75% to 60% of their Adjusted Net Bank Credit (ANBC). Hence, statement 2 is correct.
- **Inclusion of NCDC for Rural Credit:** Loans provided by banks to the National Cooperative Development Corporation (NCDC) for the purpose of on-lending to cooperative societies are now officially classified as PSL.

Other

- **Banks are also permitted to enter into co-lending arrangements to meet PSL targets.**
- **Reserve Bank of India (RBI) actually encourages banks to co-lend with NBFCs (Non-Banking Financial Companies) under the Co-Lending Model (CLM) to meet Priority Sector Lending (PSL) targets, combining bank low-cost funds with NBFC reach for underserved sectors. Hence, statement 1 is not correct.**
- **Allowed banks to treat export credit to agriculture and MSMEs as PSL loans. Hence, statement 3 is correct.**

Targets for PSL for Different Types of Banks			
Categories	Domestic commercial banks & foreign banks with 20 branches and above	Foreign banks with less than 20 branches	Regional Rural Banks
Provisions	40% of ANBC or Credit Equivalent of Off-Balance Sheet Exposures (CEOBE), whichever is higher	Same as Domestic commercial bank (32% can be in the form of Export Credit and not less than 8% can be to any other priority sector)	75% of ANBC or CEOBE whichever is higher. Lending by to Medium Enterprises, Social Infrastructure, and Renewable Energy is now capped at 15% of ANBC.

- Q 15. C** • **Recent Context:** Researchers from Birbal Sahni Institute of Palaeosciences, reconstructed climate records from peninsular India through sediment analysis of Kondagai Inland lake.
- Located in Tamil Nadu's Sivaganga district, the lake lies close to Keeladi.

About Keeladi

- **Located about 12 km southeast of Madurai, it is famous for unearthing evidence of a sophisticated urban civilization from Sangam period.**
- **Bound by river Vaigai in North. Hence, option (c) is the correct answer.**

Key Findings of the Study

- Identified three major climatic phases over Late Holocene having direct links to monsoon variability, lake hydrology and human activity in the region.
- Findings could help improve weather forecasting; disaster mapping; wetland and lake restoration efforts, etc.

- Q 16. B** • **Statement 1 is not correct:** The transfer of heat by radiation does not require any medium. It can take place whether a medium is present or not. When we come out in the sun, we feel warm. How does the heat from the sun reach us? It cannot reach us by conduction or convection as there is no medium such as air in most parts of the space between the earth and the sun. From the sun the heat comes to us by another process known as radiation.

- **Statement 2 is correct:** Since the primary source of infrared radiation is heat or thermal radiation, any object which has a temperature radiates in the infrared. Even objects that we think of as being very cold, such as an ice cube, emit infrared. When an object is not quite hot enough to radiate visible light, it will emit most of its energy in the infrared. For example, hot charcoal may not give off light but it does emit infrared radiation which we feel as heat. The warmer the object, the more infrared radiation it emits.

Q 17. D • Recent Context: Coconut is under threat from root wilt disease.

About Root Wilt Disease

- It is a debilitating, slow-spreading disease of coconut palms caused by phytoplasmas, characterized by flaccid, inwardly curving leaflets (ribbing), yellowing, root decay, and reduced yield. Hence, option (d) is the correct answer.
- Phytoplasmas are phloem-restricted plant-pathogenic bacteria transmitted by insects.

About Coconut

- Climate: Prefers humid, tropical climates, typically in coastal areas
- Temperature: 20°C to 32°C.
- Rainfall: 1000 mm is sufficient.
- It requires plenty of sunlight.
- Soil: Red sandy loam, laterite and alluvial soils
- India is the third largest coconut-producing country. (2021-22)
- Major Producing States: Kerala, Karnataka, Tamil Nadu etc.

Q 18. C Pancreas

The pancreas acts as a unique dual-function organ, with over 90% of its cells operating as an exocrine gland to aid digestion and a small percentage acting as an endocrine gland for metabolic regulation. It releases digestive enzymes (amylase, lipase) into the small intestine and secretes vital hormones (insulin, glucagon) directly into the bloodstream.

- Endocrine Function: The pancreas contains specialized clusters of cells called the Islets of Langerhans, which secrete hormones like insulin and glucagon directly into the bloodstream to regulate blood sugar levels.
- Exocrine Function: The majority (about 99%) of the pancreas consists of exocrine tissue, which produces digestive enzymes (e.g., amylase, lipase, trypsin) and an alkaline fluid (bicarbonate). These secretions are released through ducts into the small intestine (duodenum) to aid in the digestion of carbohydrates, fats, and proteins.

Hence, option (c) is the correct answer.

Q 19. A • Not all bacteria are pathogenic. While some bacteria cause diseases, the vast majority are either harmless or beneficial to humans, plants, and the environment. Many bacteria are essential for life, aiding in digestion, nutrient absorption, and food production. **Hence statement 1 is not correct.**

- Pathogens are microorganisms or agents, commonly known as germs, capable of causing infectious diseases in a host by invading the body and utilizing its resources to replicate. The primary types include viruses, bacteria, fungi, and protozoa, which can spread via bodily fluids, skin contact, airborne particles, or contaminated surfaces. **Hence statement 2 is correct.**

- Archaea are microorganisms that are generally not known to cause diseases in humans. They are often found in extreme environments and are not considered human pathogens. Archaea, particularly methanogens like *Methanobrevibacter smithii*, are common and sometimes abundant, permanent residents of the human gastrointestinal tract and oral cavity, where they behave as harmless commensals or beneficial microbes (e.g., helping with digestion). **Hence statement 3 is not correct.**

Q 20. D • Recent Context: Union Ministry of Environment, Forest, and Climate Change has notified Greenhouse Gases Emission Intensity Target (Amendment) Rules, 2025 under the Environment (Protection) Act, 1986.

Key Details

Expansion: Addition of four new sectors - Petroleum Refinery, Petrochemicals, Textiles, and Secondary Aluminium, to the list of obligated sectors to cut GHG emissions.

- Previous sectors - Aluminium, Cement, Chlor-alkali, and Pulp & Paper.
- Targets & Timeline: The new rules mandate 208 specific industrial units to reduce their GHG emission intensity (emissions per unit of product) starting from the compliance year 2025-26.
- Reduction Goals: The overall emission intensity reduction targets range from 3% to 7% by 2026-27 (baseline: 2023-24).

Compliance Mechanism: Units failing to meet targets must purchase Carbon Credit Certificates (CCCs) to cover the shortfall.

- Failure to do so attracts an 'environmental compensation' penalty equivalent to twice the average trading price of CCCs.
- Significance: This move aligns with India's Nationally Determined Contribution (NDC) target of reducing the emissions intensity of its GDP by 45% by 2030 and achieving Net Zero by 2070.

About Carbon Credit Trading Scheme (CCTS)

- **Origin: Notified by the Union Ministry of Power in 2023 under the Energy Conservation Act, 2001. Hence, statement 1 is correct.**
- Objective: CCTS laid the foundation for the Indian Carbon Market (ICM) by establishing the institutional framework.

Structure:

- **Administrator: Bureau of Energy Efficiency (BEE) sets targets and issues CCCs. Hence, statement 2 is correct.**
- Regulator: Central Electricity Regulatory Commission (CERC) regulates trading of CCCs.
- Registry: Grid Controller of India Limited.
- Operational Framework: Functions through a Compliance Mechanism (obligated entities meet GHG Emission Intensity targets; overachievers earn credits) and an Offset Mechanism (non-obligated entities voluntarily register reduction projects to seek credits).

Mechanism: It operates on a 'Cap and Trade' model.

- **Obligated entities that reduce emissions beyond their target earn CCCs (1 CCC = 1 tonne of CO₂ equivalent), which can be traded on power exchanges. Hence, statement 3 is correct.**

Q 21. C • Recent Context: Second Rangewide Estimation of Riverine and Estuarine Dolphins launched under Project Dolphin

- The previous nationwide survey (2021–23) recorded an estimated 6,327 riverine dolphins in India.
- The ongoing survey will include a new species, Irrawaddy dolphin, in Sundarbans and Odisha, apart from the Gangetic and Indus River Dolphins.

About Project Dolphin

- Launched: 15th August 2020.
- Under Centrally Sponsored Scheme: Development of Wildlife Habitats.
- Aim: To conserve both marine and riverine dolphins, along with associated cetaceans through habitat protection, scientific research, and community awareness.

River Dolphins (superfamily Platinistoidea)

- Characteristics:
 - **Functionally blind, and rely on echolocation to navigate and hunt. Hence, pair 1 is correctly matched.**
 - A long thin snout, rounded belly, stocky body and large flippers.
 - As an apex predator, they are key Indicator species of the health of rivers.
- Types of River Dolphins

Ganges river dolphin (*P. gangetica*):

- Occurs in Ganges-Brahmaputra-Meghna, and Sangu-Karnaphuli River systems in India, Bangladesh and Nepal.
- Locally called 'susu', it is endemic and India's National Aquatic Animal.
- Protection Status: Endangered (IUCN), Schedule-I (Wild Life (Protection) Act, 1972)

Indus river dolphin (*P. minor*):

- Occur in Indus basin (Beas River in India).
- **Locally called 'Bhulan', it is Punjab's State Aquatic Animal. Hence, pair 2 is correctly matched.**
- Protection Status: Endangered (IUCN), Schedule-I (WPA, 1972)

Irrawaddy dolphin (*Orcaella brevirostris*):

- Occur in marine, brackish, and freshwater environments throughout southeast Asia.
- **Slow swimmers, have unique water spitting behavior. Hence, pair 3 is correctly matched.**
- Protection Status: Endangered (IUCN)

Q 22. B Innate immunity

- It is the body's rapid, non-specific, first-line defense present from birth, using physical barriers (skin, mucus), chemical barriers (stomach acid, tears), and general cells (phagocytes, NK cells) to immediately fight off common invaders like bacteria and viruses, preventing infection and triggering the adaptive immune system for targeted responses.
- Key components include physical barriers like the skin and mucous membranes, as well as specialized cells (e.g., macrophages, neutrophils, natural killer cells) and proteins (e.g., the complement system, interferons) that work to identify and eliminate threats.
- Unlike the adaptive immune system, the innate immune system does not have an immunological memory, meaning it cannot provide long-lasting, enhanced protection against future encounters with a specific pathogen.

Hence, option (b) is the correct answer.

- Q 23. B • **Recent Context:** A powerful winter storm has paralysed the Kamchatka Peninsula in Russia's Far East, burying streets and homes under massive amounts of snow.
- **The Kamchatka Peninsula is a 1,250-kilometre-long peninsula in the Russian Far East. It lies between the Pacific Ocean to the east and the Sea of Okhotsk to the west. It is famous for its high density of volcanoes. Hence, pair 1 is correctly matched.**



- **US President Trump has criticised the potential return of Diego Garcia by the UK to Mauritius, which hosts a vital US military base. Diego Garcia is an atoll located south of the equator in the central Indian Ocean. Hence, pair 2 is correctly matched.**



- The Negev Desert is located in Southern Israel (West Asia/Middle East), not South America. It covers over half of Israel's total land area. It is frequently in the news due to its strategic importance and the development of solar energy and desert agriculture. Hence, pair 3 is not correctly matched.



- Q 24. A
- A **fluorescent lamp**, or fluorescent tube, is a **low-pressure mercury-vapour gas-discharge lamp** that uses fluorescence to produce visible light.
 - An electric current in the gas excites mercury vapour, which produces short-wave ultraviolet light that then causes a phosphor coating on the inside of the lamp to glow. **Hence option (a) is the correct answer.**

- **Astatine** is a radioactive element of the halogen series: a decay product of uranium and thorium that occurs naturally in minute amounts and is artificially produced by bombarding bismuth with alpha particles. There are currently no uses for astatine outside of research.
- **Cinnabar** is a toxic mercury sulfide mineral with a chemical composition of HgS. It is the only important ore of mercury. It has a bright red colour that has caused people to use it as a pigment, and carve it into jewellery and ornaments for thousands of years in many parts of the world.

Q 25. C • The Bletchley Declaration, recently seen in the news, is associated with the **world's first Artificial Intelligence (AI) Safety Summit held at Bletchley Park in Buckinghamshire, near London**. The summit, spanning November 1-2, **brought together global leaders, computer scientists, and tech executives to discuss and address the challenges and risks posed by AI, particularly focusing on "frontier AI."**

- The summit aimed to establish a framework for mitigating the risks **associated with AI while maximizing its potential**. On the first day of the summit, a groundbreaking agreement, known as the **Bletchley Declaration**, was reached. This declaration involves a **commitment from 28 major countries, including the United States, China, Japan, the United Kingdom, France, India, and the European Union**.
- The Bletchley Declaration defines **"frontier AI" as highly capable foundation generative AI models** with the potential for dangerous capabilities that pose severe risks to public safety.
- The Bletchley Declaration underscores that the **risks associated with AI**, including bias and privacy concerns, are best addressed through international cooperation. It emphasizes the importance of collaboration among countries to develop a shared understanding of both the opportunities and risks posed by frontier AI.
- As part of the agreement on international collaboration, **South Korea** is set to co-host a virtual AI summit in the next six months. Additionally, **France will host the next in-person summit** within a year.
- **Hence option (c) is the correct answer.**

Q 26. A • Monroe doctrine

- Recent policies of the present US President Donald Trump are referred to as The 'Donroe doctrine' as these largely reflect the 200 years old 'Monroe Doctrine'.

About Monroe doctrine

- Proclaimed: In 1823 by U.S. President James Monroe (United States' policy on the new political order).
- **Objective: To establish U.S. dominance and security in its surrounding region. Hence, option (a) is the correct answer.**

The three main concepts of the doctrine

- **Non-colonization:** European powers should not establish new colonies in the Americas.
- **Non-intervention:** Europe must not interfere in the affairs of independent American states.
- **U.S. Non-interference:** The U.S. would not meddle in European wars or colonies.

Q 27. C • Redox reaction is a short name for oxidation-reduction reactions. As the name implies, it involves two interdependent half reactions, **oxidation** and **reduction**. Oxidation refers to when the reductant

loses electrons to the oxidant and itself gets oxidized. Reduction refers to when the oxidant gains electrons from the reductant and itself gets reduced. Oxidation and reduction need to take place at the same time.

- **Hence option (c) is the correct answer.**

Q 28. C • Water is capable of dissolving a variety of different substances, which is why it is such a good solvent. And, water is called the "universal solvent" because it dissolves more substances than any other liquid. This is important to every living thing on earth. It means that wherever water goes, either through the ground or through our bodies, it takes along valuable chemicals, minerals, and nutrients.

- **It is water's chemical composition and physical attributes that make it such an excellent solvent. Water molecules have a polar arrangement of the oxygen and hydrogen atoms—one side (hydrogen) has a positive electrical charge and the other side (oxygen) had a negative charge.** This allows the water molecule to become attracted to many other different types of molecules. Water can become so heavily attracted to a different molecule, like salt (NaCl), that it can disrupt the attractive forces that hold the sodium and chloride in the salt molecule together and, thus, dissolve it.

- **Hence option (c) is the correct answer.**

Q 29. D • **Recent Context:** Prime Minister recently inaugurated the Kaziranga Elevated Corridor Project designed to ensure safe wildlife movement across the Kaziranga National Park.

About Kaziranga National Park (KNP)

- Location: Between the Brahmaputra River and the Karbi (Mikir) Hills in Assam.
- **Status: National Park, Tiger Reserve, Important Bird Area (IBA), and UNESCO World Heritage Site (1985). Hence, option (d) is the correct answer.**
- An important area for migratory birds as it is located at the junction of the Australasia and Indo-Asian flyway.
- Biodiversity found: Tigers, elephants, swamp deer, wild buffalo, etc.
- Largest population of Indian one-horned rhinoceros found here.

Q 30. B • **Isotopes are defined as the atoms of the same element, having the same atomic number but different mass numbers. Hydrogen atom has three atomic species, namely protium (1 H), deuterium (2 H or D) and tritium (3 H or T). The atomic number of each one is 1, but the mass number is 1, 2 and 3, respectively. Hence option (b) is the correct answer.**

- While Isobar is defined as , atoms with identical atomic number but different atomic mass number for e.g. two elements — calcium, atomic number 20, and argon, atomic number 18 but the mass number of both these elements is 40.

Q 31. C Carbohydrates such as glucose are the body's primary and immediate source of energy. Polysaccharides like glycogen (in animals) and starch (in plants) serve as important reserve sources of energy. **Hence, statement 1 is correct.**

The majority of enzymes are globular proteins. However, some ribonucleic acid (RNA) molecules also possess catalytic activity. These are called ribozymes, and they are not proteins. **Hence, statement 2 is not correct.**

The human gut microbiota synthesises essential vitamins, such as Vitamin K and some B vitamins (e.g., B12, biotin, folate), which are then absorbed and used by the body for metabolism. **Hence, statement 3 is correct.**

- Q 32. A** • An **Einstein ring** is a ring of light around a form of **dark matter**, galaxy, or cluster of galaxies. It is an example of gravitational lensing. **Gravitational lensing** is a phenomenon that occurs when a massive celestial body — such as a galaxy or cluster of galaxies — creates a gravitational field that distorts and amplifies the light from distant galaxies that are behind it but in the same line of sight. The body causing the light to curve is called a gravitational lens. **Hence statement 1 is correct.**
- Einstein's rings are named after mathematician and physicist **Albert Einstein**, whose **general theory of relativity** predicted that light could bend and brighten around objects across the cosmos. The **first Einstein ring was discovered in 1987**, and several more have been discovered since then. Recently, the **European Space Agency's (ESA) Euclid space telescope** has discovered an Einstein ring around a galaxy nearly 590 million light-years away from Earth. The ring was discovered around **NGC 6505**, a galaxy first found in the 19th Century. **Hence statement 2 is not correct.**

- Q 33. A** • **The twinkling of a star is due to atmospheric refraction of starlight.** The starlight, on entering the earth's atmosphere, undergoes refraction continuously before it reaches the earth. The **atmospheric refraction occurs in a medium of gradually changing refractive index.** Since the atmosphere bends starlight towards the normal, the apparent position of the star is slightly different from its actual position. The star appears slightly higher (above) than its actual position when viewed near the horizon. **Hence option (a) is the correct answer.**
- Further, this apparent position of the star is not stationary, but keeps on changing slightly, since the physical conditions of the earth's atmosphere are not stationary. Since the stars are very distant, they approximate point-sized sources of light. As the path of rays of light coming from the star goes on varying slightly, the apparent position of the **star fluctuates and the amount of starlight entering the eye flickers** – the star sometimes appears brighter, and at some other time, fainter, which is the **twinkling effect.**

- Q 34. C** • **Friction is the resistive force acting between bodies that tends to oppose and damp out motion.**
- Friction is usually distinguished as being either **static friction (the frictional force opposing placing a body at rest into motion)** and kinetic friction (the frictional force tending to slow a body in motion). In general, static friction is greater than kinetic friction.
 - Friction always works in the direction opposite to the direction in which the object is moving, or trying to move.

Friction is caused by the irregularities on the two surfaces in contact.

- Even those surfaces which appear very smooth have a large number of minute irregularities on them. Irregularities on the two surfaces lock into one another.
- When we attempt to move any surface, we have to apply a force to overcome interlocking. On rough surfaces, there are a larger number of irregularities.
- **So the force of friction is greater if a rough surface is involved.**

- Q 35. C • Conventionally, in an electric circuit, the direction of electric current (both AC and DC) is taken as opposed to the direction of the flow of electrons, which are negative charges. **Hence statement 1 is not correct.**
- In direct current (DC), the electric charge (current) only flows in one direction. Electric charge in alternating current (AC), on the other hand, changes direction periodically. **Hence statement 2 is correct.**
 - A battery produces only DC power. Generators are required for producing AC power. **Hence statement 3 is correct.**
- Q 36. B • **Small modular reactors (SMRs)** are advanced nuclear reactors with a power capacity of up to **300 MW(e) per unit**, about **one-third of the generation capacity of traditional nuclear power reactors**. Small-capacity atomic power plants, popularly called Small Modular Reactors (SMRs), with their unique features of modularity, scalability, small footprint, and improved safety, present themselves as an attractive option for repurposing retiring coal-based thermal power station sites. **Hence statement 1 is not correct.**
- Many of the benefits of SMRs are inherently linked to the nature of their design – small and modular. Given their smaller footprint, SMRs can be sited in locations unsuitable for larger nuclear power plants. **Prefabricated units of SMRs can be manufactured, shipped, and installed on-site, making them more affordable to build than large power reactors**, which are often custom-designed for a particular location, sometimes leading to construction delays. SMRs offer savings in cost and construction time and can be deployed incrementally to match increasing energy demand. **Hence statement 2 is correct.**
 - Compared to existing nuclear reactors, SMR designs are generally simpler, and the safety concept for SMRs often relies more on passive systems and inherent safety characteristics of the reactor, such as low power and operating pressure. This means that in such cases, **no human intervention or external power or force is required to shut down systems because passive systems rely on physical phenomena, such as natural circulation, convection, gravity, and self-pressurization**. These increased safety margins, in some cases, eliminate or significantly lower the potential for unsafe releases of radioactivity to the environment and the public in case of an accident. **Hence statement 3 is correct.**
 - **Hence option (b) is the correct answer.**
- Q 37. B • In chemistry, reactivity is a measure of how readily a substance undergoes a chemical reaction. The reaction can involve the substance on its own or with other atoms or compounds, generally accompanied by a release of energy.
- In general, the more reactive a metal is: the more vigorously it reacts with other substances the more easily it loses electrons to form positive ions (cations) We can examine the reactivity of metals by observing their reactions with oxygen, water, steam and whether it displaces other metals in displacement reactions.
 - **Relative reactivities of metals**

K	Potassium	Most reactive
Na	Sodium	
Ca	Calcium	
Mg	Magnesium	
Al	Aluminium	
Zn	Zinc	Reactivity decreases
Fe	Iron	
Pb	Lead	
H	Hydrogen	
Cu	Copper	
Hg	Mercury	
Ag	Silver	
Au	Gold	Least reactive

- Hence, option (b) is the correct answer.

Q 38. A • **Statement 1 is correct:** Sound is a mechanical wave and needs a material medium like air, water, steel etc. for its propagation. It cannot travel through vacuum.

- **Statement 2 is not correct:** In these waves the individual particles of the medium move in a direction parallel to the direction of propagation of the disturbance. The particles do not move from one place to another but they simply oscillate back and forth about their position of rest. This is exactly how a sound wave propagates, hence sound waves are longitudinal waves.

- There is also another type of wave, called a transverse wave. In a transverse wave particles do not oscillate along the direction of wave propagation but oscillate up and down about their mean position as the wave travels. Thus, a transverse wave is the one in which the individual particles of the medium move about their mean positions in a direction perpendicular to the direction of wave propagation.

- **Statement 3 is not correct:** The speed of sound remains almost the same for all frequencies in a given medium under the same physical conditions.

Q 39. C • A colloidal system is a heterogeneous mixture where particles of one substance (dispersed phase) are distributed within another substance (dispersion medium). The classification depends on the state of the dispersed phase and dispersion medium.

Sol

- A sol is a colloid where solid particles are dispersed in a liquid.
- Example: Paint, blood, ink.

Emulsion

- An emulsion is a colloid where both the dispersed phase and dispersion medium are liquids.
- Example: Milk, butter, mayonnaise.

Aerosol

- An aerosol is a colloid where liquid or solid particles are dispersed in a gas.
- Example: Fog, mist, clouds (liquid in gas) and smoke, dust (solid in gas).
- Hence, option (c) is the correct answer.

Foam

- A foam is a colloid where a gas is dispersed in a liquid or solid.
- Example: Soap lather (gas in liquid), sponge (gas in solid).

- Q 40. C • Recent Context:** Ministry of Science and Technology recommended the use of steel slag-based technology for sustainable road construction, particularly in hilly terrains.

Ministry's Technology Development Board (TDB) also signed an agreement for commercial rollout of 'ECOFIX'.

- It is a pothole repair mix developed by CSIR–Central Road Research Institute (CRRI). Hence, option (c) is the correct answer.

Steel slag

- Steel slag is a solid waste generated in the process of steel making and is mainly composed of oxides of calcium, iron, silicon, magnesium, and some other elements.
- Properties: Possesses strength, durability, and corrosion resistance.
- Applications: Suitable for use as a construction material, concrete and cement applications, road construction etc.

- Q 41. B • All the options are correct.**

- **Doppler Effect:** The Doppler effect is a phenomenon observed whenever the source of waves is moving with respect to an observer. The **Doppler effect** can be described as the effect produced by a moving source of waves in which there is an apparent upward shift in frequency for the observer and the source are approaching and an apparent downward shift in frequency when the observer and the source are receding. The Doppler effect can be observed to occur with all types of waves - most notably water waves, sound waves, and light waves.
- **Examples:** A police car or emergency vehicle was traveling towards you on the highway. As the car approached with its siren blasting, the pitch of the siren sound (a measure of the siren's frequency) was high; and then suddenly after the car passed by, the pitch of the siren sound was low. That was the Doppler effect - a shift in the apparent frequency for a sound wave produced by a moving source.

Applications of Doppler Effect:

- Police radar to check the speed of vehicles on a highway.
- To read weather events. In this case, the stationary transmitter is located in a weather station and the moving object being studied is a storm system.
- Echocardiogram.
- **Sonic boom produced by the supersonic aircraft.**
- Echolocation by bats: To help them find their prey in the dark, most bat species have developed a remarkable navigation system called **echolocation**

- Q 42. A • Recent Context:** Prime Minister participated in “Bagurumba Dwhou 2026”, celebrating rich heritage of the Bodo community in Assam.

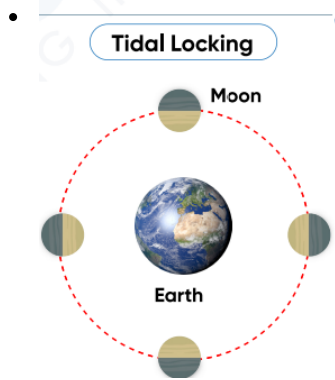
About Bagurumba Dance

- **Traditional folk dance of the Bodo community of Assam. Hence, statement 1 is correct.**
- Inspired by nature, symbolises blooming flowers, reflects harmony between human life and the natural world.
- **Traditionally performed by young Bodo women, men accompany as musicians, playing traditional instruments.**

- Usually performed in groups, forming circles or lines that enhance its visual elegance. Hence, statement 2 is not correct.
- Cultural Significance: Dance represents peace, fertility, joy and collective harmony, and is closely associated with festivals such as Bwisagu, the Bodo New Year, and Domasi.

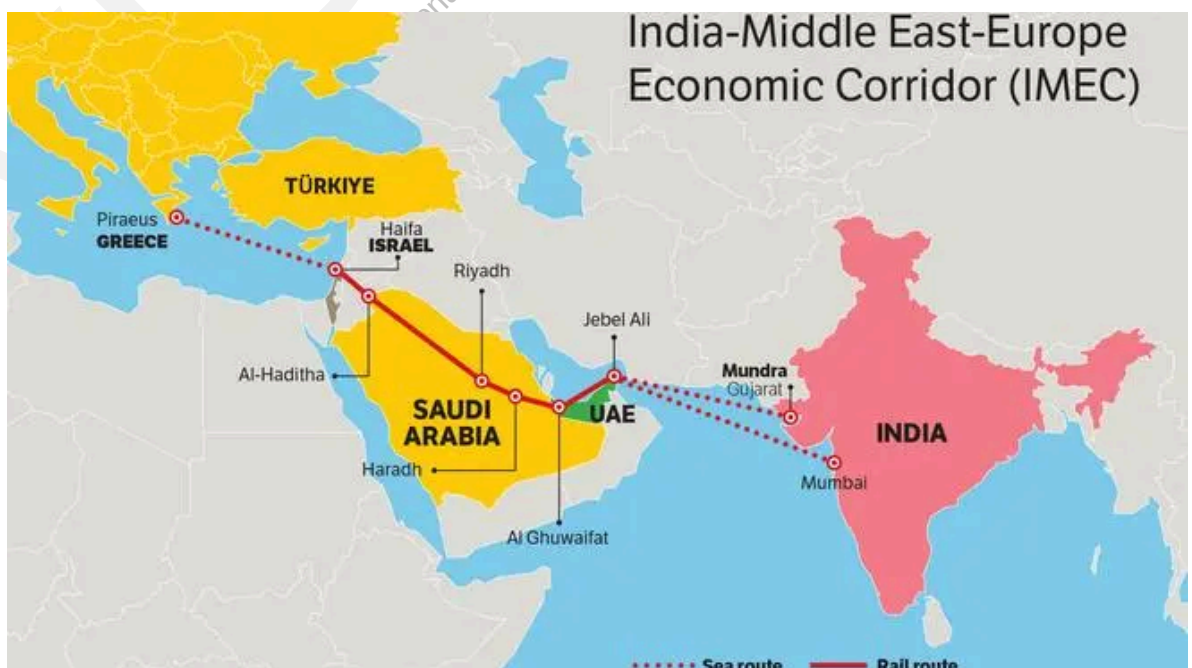
- Q 43. C**
- Resonance is a phenomenon in which a vibrating system or external force drives another system to oscillate with greater amplitude at a specific preferential frequency. Frequencies at which the response amplitude is a relative maximum are known as the system's resonant frequencies or resonance frequencies.
 - A highly trained singer can hit a note that causes glass to shatter due to the phenomenon of resonance. The natural, or resonant, frequency in the glass, as with all objects, is determined by its shape and composition. If the singer's voice (or a note from an instrument) hits the resonant frequency, there will be a transfer of energy. A full transfer of energy from the voice or musical instrument can overload the glass, causing it to shatter.
 - **Interference of Sound:** Two traveling waves that exist in the same medium will interfere with each other. If their amplitudes add, the interference is said to be constructive interference, and destructive interference if they are "out of phase" and subtract.
 - **Reverberation** is a result of multiple reflections. A sound wave in an enclosed or semi-enclosed environment will be broken up as it is bounced back and forth among the reflecting surfaces. Reverberation is, in effect, a multiplicity of ECHOes whose speed of repetition is too quick for them to be perceived as separate from one another.
 - The behavior of a wave (or pulse) upon reaching the end of a medium is referred to as **boundary behavior**. There are essentially four possible behaviors that a wave could exhibit at a boundary: reflection (the bouncing off of the boundary), diffraction (the bending around the obstacle without crossing over the boundary), transmission (the crossing of the boundary into the new material or obstacle), and refraction (occurs along with transmission and is characterized by the subsequent change in speed and direction).
 - Hence option (c) is the correct answer.

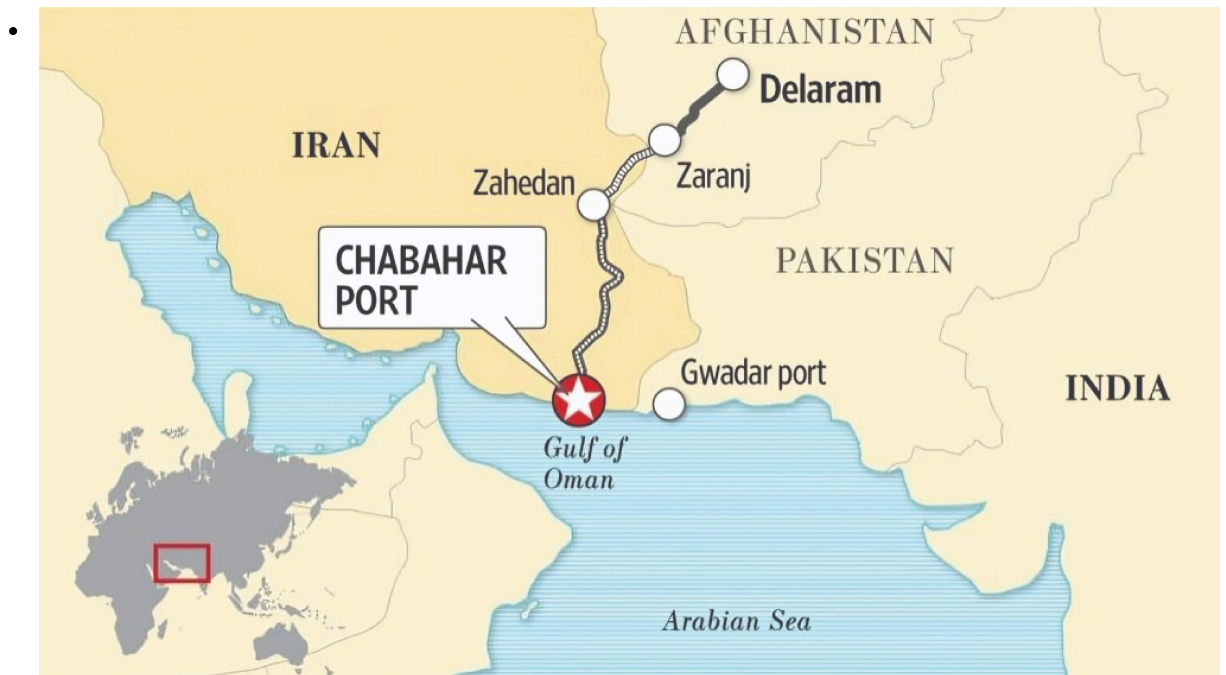
- Q 44. B**
- As the sun rises on the Moon, ISRO is trying to wake up rover Pragyan and lander Vikram which were put to sleep to survive the lunar night.
 - It refers to the synchronization between the Moon's rotation around its axis (with respect to the Sun) and its orbit around the Earth. It is due to the gravitational force exerted by both the Moon and the Earth on each other. All the solar system's large moons are tidally locked with their planets. Some binary stars are tidally locked to one another. Hence statement 1 is correct.
 - The effect arises between two bodies when **their gravitational interaction slows a body's rotation until it becomes tidally locked**. Over many millions of years, the interaction forces changes to their orbits and rotation rates as a result of energy exchange and heat dissipation. When one of the bodies reaches a state where there is no longer any net change in its rotation rate over the course of a complete orbit, it is said to be tidally locked. Hence statement 2 is not correct.



- **Tidal locking causes bodies to stretch and distort, leading to a deformation of the Moon's shape.** Additionally, Earth's gravitational force on the Moon results in alternating high and low ocean tides. **Hence statement 3 is correct.**
- Tidal locking between the **Moon and Earth results in the Moon always presenting the same face to Earth.** This leads to the emergence of two distinct sides of the Moon. The Near Side, visible from Earth, comprises about 60% of the Moon's surface and features a relatively smoother, thinner crust with large volcanic plains.
- On the other hand, the **Far Side, never visible from Earth, is illuminated in sunlight during the 'new moon' phase,** receiving light for nearly a fortnight. Despite being hidden from direct view, the Far Side is not permanently dark; it merely conceals its various topographical features during this tidal locking-induced phenomenon.
- **Hence option (b) is the correct answer.**

- Q 45. A**
- Recent Context: India is exploring options to retain presence in Iran's Chabahar port amid US pressure on Iran.
 - Port of Piraeus (Greece, Mediterranean) — westernmost
 - Haifa Port (Israel, eastern Mediterranean) — east of Piraeus
 - Jebel Ali Port (UAE, Persian Gulf) — further east
 - Chabahar Port (Iran, Gulf of Oman) — easternmost among these.
 - **Hence, option (a) is the correct answer.**



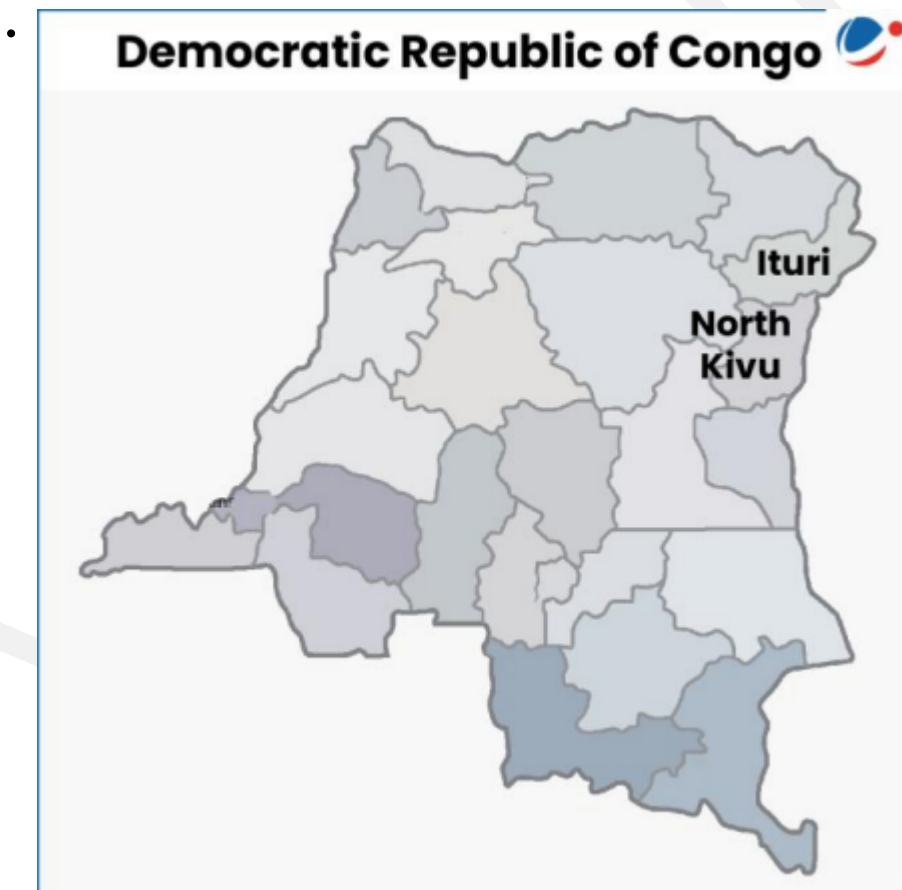


- Q 46. B** **The TSAT-1A is India's first private sector-built military-grade geospatial satellite.**
- It's an optical satellite that delivers high-resolution images, and is expected to revolutionize Earth observation. **Hence, statement 1 is correct.**
 - The satellite weighs 50 kg and is placed in low earth orbit.
 - It's capable of increased collection capacity, dynamic range, and low-latency delivery.
 - **It has been successfully deployed into space by SpaceX's Falcon 9 rocket. Hence, statement 2 is not correct.**
 - **It was developed by Tata Advanced Systems Limited (TASL) in collaboration with Satellogic Inc.**
- Key Features:**
- Sub-meter resolution for capturing detailed images of Earth's surface.
 - Multispectral and hyperspectral imaging capabilities for collecting data across a wide range of wavelengths.
 - Greater collection capacity, wider dynamic range, and low-latency delivery of data.
- It carried into space as part of the Bandwagon-1 rideshare mission. Hence, statement 3 is correct.**
- **Bandwagon-1 mission is a rideshare mission organized by SpaceX to launch multiple small satellites into orbit.**
 - It is SpaceX's first dedicated rideshare mission launched into a mid-inclination orbit.
 - Other Mission: Alongside TSAT-1A, the Bandwagon-1 mission had 11 other spacecraft.
- Q 47. D** • **Capillary action** is defined as the spontaneous flow of a liquid into a narrow tube or porous material.
- This **movement does not require the force of gravity** to occur. It often **acts in opposition to gravity**.
 - Capillary action is sometimes called capillary motion, capillarity, or wicking. Capillary action is caused by the combination of cohesive forces of the liquid and the adhesive forces between the

liquid and tube material.

- **Cohesion and adhesion** are two types of intermolecular forces. These forces pull the liquid into the tube. In order for wicking to occur, a tube needs to be sufficiently small in diameter.
- Examples of capillary action include:
- Capillary action in plants is one of the most essential processes which helps them to survive. The **roots of plants draw water via capillary action** from the soil, and thus, supply it to various parts of the plant including the stems, branches, and leaves. **Hence option 3 is correct.**
- **A candle burns due to capillary action.** When the wick of a candle is lit, the heat of the flame causes the candle wax to melt. The molten wax is then pulled up through the wick by capillary action and supplies the fuel needed to sustain the fire and keep the candle burning. **Hence option 2 is correct.**
- **Pores in the sponge act as minute capillaries**, causing them to absorb a large number of liquids. **Hence option 1 is correct.**
- Other examples include uptake of **water in paper and plaster** (two porous materials), the **movement of water through sand** etc.

Q 48. A • Ituri is a province in the Democratic Republic of the Congo (DRC), known for ethnic conflict and humanitarian crises. **Hence, pair 1 is not correctly matched.**



- Khartoum is the capital of Sudan and has been a major focus of news due to the ongoing civil war between the SAF and RSF. **Hence, pair 2 is correctly matched.**



- Chernihiv is a historic city and oblast (region) in Ukraine, frequently in the news due to its proximity to the Russian border. **Hence, pair 3 is not correctly matched.**



- Q 49. C**
- The effects of forces of the same magnitude on different areas are different. When the person stands on loose sand, the force, that is, the weight of his body is acting on an area equal to the area of his feet. When he lies down, the same force acts on an area equal to the contact area of his whole body, which is larger than the area of his feet. In both cases, the force exerted on the sand is the weight of his body and the force is acting perpendicular to the surface of the sand. The force acting on an object perpendicular to the surface is called thrust. The effect of thrust on sand is larger while standing than while lying. The thrust on the unit area is called pressure.
 - $\text{Pressure} = \text{Thrust} / \text{Area}$
 - **Thus, higher pressure is exerted by the body on the sand while standing and the person goes down deeper into it as compared to when he is lying down.**

- Q 50. B** • The Indian Army initiated Project Sanjay with the aim of transforming itself into a future-ready, technology-driven, lethal, and agile force. The project focuses on building the Army's Battlefield Surveillance System.

Key Objectives:

- The primary objective of this system is to provide commanders and staffers with a composite operational picture for faster decision-making.
- It integrates ground-based sensors, UAV feed, and satellite data that is needed for tactical battle and to look at depth areas that are 200 km beyond the boundaries.

Implementation

- Bharat Electronics Limited is the system integrator of this ambitious project.
 - It has validated the project in various terrains and is now delivering the surveillance centers for field formations by December 2025.
 - The information and data from thousands of secure sensors will be combined into a comprehensive system that will help the Army commanders with quick decision-making capabilities.
- **Other Projects** The Army is also working on several other projects, including SAMA, Combat Information Decision Support System, E-Sitrep, Project Avgat, Project Anumaan, and Project INDRA. These projects aim to strengthen the force by achieving secure information dominance in all spheres of its work, including the operational, administrative, and managerial spheres. **Hence option (b) is the correct answer.**

- Q 51. D** • Prokaryotic cells are simple, single-celled microorganisms from the domains Bacteria and Archaea that lack a membrane-bound nucleus and organelles. They possess a cell wall, plasma membrane, cytoplasm, ribosomes, and a single, circular chromosome located in a nucleoid region. They reproduce primarily by binary fission. **Hence, statement 1 is correct.**
- Prokaryotic organisms (bacteria and archaea) lack a nucleus and true membrane-bound organelles, meaning their genetic material (usually a single, circular chromosome) is not enclosed within a nuclear membrane. Instead, the DNA resides in a centralised, unbound region of the cytoplasm known as the nucleoid. **Hence, statement 2 is correct.**
 - Some bacteria, which are prokaryotes, are obligate intracellular parasites. An example is the genus Chlamydia, which must live and reproduce inside host cells. **Hence, statement 3 is correct.**

- Q 52. A** • Bioluminescence is the natural production and emission of light by a living organism through a chemical reaction (luciferin + oxygen, catalyzed by luciferase), creating "cold light" with little heat, seen in fireflies, deep-sea creatures, fungi, and bacteria. Thus during bioluminescence, **luciferin** is oxidised (via **luciferase**) to **oxyluciferin**, releasing energy as visible light. **Hence option (a) is the correct answer.**

- Q 53. C** • A **plane mirror** is a mirror with a flat and smooth reflecting surface. Reflection from a smooth surface like that of a mirror is called regular reflection. Images are formed by regular reflection.

Some properties of images formed by plane mirror are as follows:

- **The image formed by a plane mirror is erect.**

- It is virtual (cannot be obtained on a screen) and is of the same size as the object. This is because the light rays do not actually pass through the image. So, it cannot be focused on a screen placed at the location where the image is.
- **The image is at the same distance behind the mirror as the object is in front of it.**
- The left side of the object is seen on the right side in the image, and right side of the object appears to be on the left side in the image. This is known as **lateral inversion**.

Q 54. C • Recent Context: Cabinet approves continuation of APY till 2030-31.

About APY:

- Launch: 9th May, 2015
- Objective: To create a universal social security system for all, especially the poor, underprivileged and workers in the unorganized sector.
- **Eligibility: Any citizen of India between the age of 18 and 40 who are not income tax payers. Hence, statement 2 is not correct.**
- **Type: Central Sector Scheme. Hence, statement 1 is correct.**
- **Scheme Features: APY offers a guaranteed minimum pension of Rs.1,000 to Rs.5,000 per month starting at age 60, based on contributions. Hence, statement 3 is correct.**
- Progress: Over 8.66 crore subscribers have been enrolled, making APY a cornerstone of India's inclusive social security framework.

Q 55. C • Recent Context: Union Home Minister paid tribute to freedom fighter Ras Behari Bose on his death anniversary.

About Rash Behari Bose (1886–1945)

- Born in Bardhaman district (Bengal).
- Was deeply inspired by the French Revolution of 1789.

Key Contributions

- Was an active member of the Yugantar group of revolutionaries under the leadership of Motilal Roy.
- Acted as an effective link between revolutionaries of Punjab, United Provinces and Bengal.
- **Involved in the Delhi Conspiracy Case (1912) related to the bomb attack on Viceroy Lord Hardinge.**
- **Founded the Indian Independence League (1942) in Tokyo.**
- **Played a key role in the Ghadar Movement and in the formation of the Azad Hind Fauj (Indian National Army). Hence, option (c) is the correct answer.**
- Japanese Government honoured him with the Order of the Rising Sun (2nd grade)

Q 56. D • Transcription is the foundational molecular process where RNA polymerase enzymes catalyze the synthesis of an RNA strand by reading a specific DNA template strand. As the first step in gene expression, this process ensures genetic information is copied into RNA, which is later used to create proteins. **Hence, statement 1 is correct.**

- In eukaryotic cells, transcription—the synthesis of RNA from a DNA template—occurs exclusively within the membrane-bound nucleus. This compartmentalization allows for post-transcriptional

modifications, such as mRNA splicing and capping, before the transcript moves to the cytoplasm for translation. Three types of RNA polymerase (I, II, and III) carry out this process. **Hence, statement 2 is correct.**

- RNA polymerase is the essential multi-unit enzyme responsible for transcription, the process of copying DNA into RNA. It binds to promoter sequences to initiate, elongates the RNA chain using a single-stranded DNA template, and terminates synthesis. Prokaryotes use one type, while eukaryotes use three distinct types (I, II, III). **Hence, statement 3 is correct.**

Q 57. B • A decomposition reaction can be defined as a chemical reaction in which one reactant breaks down into two or more products. Burning of coal requires a combination of carbon and oxygen, which releases carbon dioxide. **Hence pair 1 is not correctly matched.**

- **When two or more substances (elements or compounds) combine to form a single product, the reactions are called combination reactions.** The formation of liquid water involves a combination of hydrogen and oxygen in the gaseous state. **Hence pair 2 is correctly matched.**

- **If a substance gains oxygen during a reaction, it is said to be oxidized. If a substance loses oxygen during a reaction, it is said to be reduced.** Corrosion is a chemical or electrochemical reaction that occurs when a metal or alloy is exposed to a corroding agent, such as oxygen in the air or water. The metal loses electrons to the oxygen, forming an oxide. **Hence pair 3 is correctly matched.**

- **Hence option (b) is the correct answer.**

Q 58. D • Recent Context: SC is once again examining the long-standing dispute over the ST status of the Lambada community in Telangana.

About Lambada Tribe

- Also known as Sugalis or Banjaras, They are one of the Scheduled Tribe communities Spread across Telangana and Andhra Pradesh
- Origin: Believed to have originated from Marwar region of Rajasthan.
- Traditional Occupation: With the advent of British rule, the Lambadas were compelled to give up their semi-nomadic traditional occupation of transporting goods.
- **Language Spoken: “Gor Boli” or Lambadi’. Hence, option (d) is the correct answer.**
- Culture: Distinct dress, rich embroidery, traditional music by Dappans.
- **Kurukh:** This is a Dravidian language spoken by the Oraon and Kisan tribes, primarily in Jharkhand, Chhattisgarh, Odisha, and West Bengal.
- **Mundari:** Belonging to the Austroasiatic language family, it is spoken by the Munda people in the Chhota Nagpur Plateau region (Jharkhand, Odisha, and Bihar). It is historically significant as the language associated with the Ulgulan (Great Tumult) led by Birsa Munda.

Q 59. C • In 2018, the definitions of the kilogram, the ampere, the kelvin and the mole were changed.

- The **kilogram’s definition** will be based on a concept of physics called the **Planck constant**. The Planck constant is a quantity that relates a light particle’s energy to its frequency.
- The new measure for kilogram from 2019 derived from constants of nature that are all interdependent. It will be set by the value of Planck constant in combination with definitions of meter and second. **Kibble balance will be used to make accurate measures of Planck constant.**

Kibble balance is a set of scales, which uses force produced by current-carrying wire in a magnetic field to balance the weight of the mass.

- Apart from Mass, there are six base quantities as per the SI (International System of Units).
- The International System of Units (SI), commonly known as the metric system, is the international standard for measurement. The International Treaty of the Meter was signed in Paris on May 20, 1875, by seventeen countries, including the United States and is now celebrated around the globe as World Metrology Day.
- The SI rests on a foundation of seven (7) defining constants: the caesium hyperfine splitting frequency, the speed of light in vacuum, the Planck constant, the elementary charge (i.e. the charge on a proton), the Boltzmann constant, the Avogadro constant, and the luminous efficacy of a specified monochromatic source. Definitions of all seven (7) SI base units are expressed using an explicit-constant formulation and experimentally realized using a specific *mises en pratique* (practical technique).

The seven SI base units, which are comprised of:

- **Length - meter (m)**
- Time - second (s)
- Amount of substance - mole (mole)
- Electric current - ampere (A)
- **Temperature - kelvin (K)**
- Luminous intensity - candela (cd)
- **Mass - kilogram (kg)**
- **Hence only options 1, 2 and 4 are correct.**

- Q 60. B**
- Tuberculosis, also called TB, is a serious illness that mainly affects the lungs. The germs that cause tuberculosis are a type of bacteria. Tuberculosis can spread when a person with the illness coughs, sneezes or sings. This can put tiny droplets with the germs into the air.
 - **Malaria is a life-threatening disease caused by Plasmodium parasites**, which are transmitted to humans through the bites of infected female Anopheles mosquitoes. Malaria is a life-threatening disease spread to humans by some types of mosquitoes. It is mostly found in tropical countries. It is preventable and curable. The infection is caused by a parasite and does not spread from person to person.
 - Plague is an infectious disease caused by the bacterium *Yersinia pestis*, primarily found in small mammals and their fleas. While famous for causing historic pandemics like the "Black Death," it remains a persistent public health threat in parts of Africa, Asia, and the Americas.
 - **Hence, option (b) is the correct answer.**

- Q 61. B**
- **India's 2026 Diplomatic Strategy: From Large Forums to Strategic Coalitions Shifting Diplomatic Landscape of 2026**
 - Demanding Bilateral relations: Bilateral diplomacy, particularly with major powers, is expected to remain "demanding," with ties to both Washington and Beijing continuing to generate friction from trade disputes and strategic competition.
 - Strain on Large Multilateral Forums: Large, established forums like UN and G20 are proving to be increasingly ineffective for concrete action due to geopolitical rivalries and the intrusion of

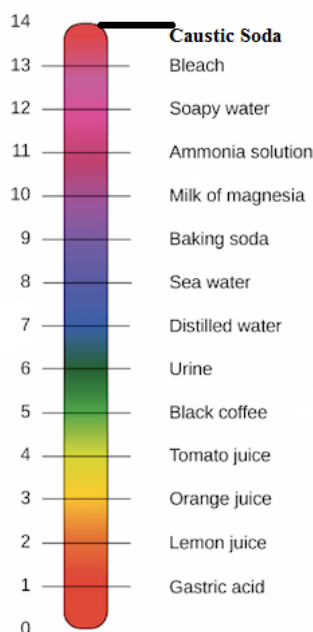
domestic politics.

- e.g., US boycott of the Johannesburg G-20 summit in 2025.

Engaging "Diplomatic White Spaces": Key Arenas for "Small Table" Diplomacy

- **Diplomatic Whitespaces are defined as "gaps in global leadership" where pressing problems require coordination, but no major power can serve as a credible convenor. Hence, option (b) is the correct answer.**

- Q 62. B**
- pH is a quantitative measure of the acidity or basicity of aqueous or other liquid solutions. The term, widely used in chemistry, biology, and agronomy, translates the values of the concentration of the hydrogen ion—which ordinarily ranges between 0 and 14 on a pH scale.
 - The strength of acids and bases depends on the number of H^+ ions and OH^- ions produced, respectively. If we take hydrochloric acid and acetic acid of the same concentration, say one molar, then these produce different amounts of hydrogen ions. Acids that give rise to more H^+ ions are said to be strong acids, and acids that give less H^+ ions are said to be weak acids.



Approximate pH values of some common acids and bases are

- **Lemon juice** - 2.2
- **Milk of Magnesia (Magnesium hydroxide)** - 10
- **Caustic Soda (Sodium Hydroxide)** - 14
- Water has a pH of around 7.4.
- Our body works within the pH range of 7.0 to 7.8. Living organisms can survive only in a narrow range of pH changes. When the pH of rainwater is less than 5.6, it is called acid rain.
- **Hence, option (b) is the correct answer.**

- Q 63. C**
- Plaster of Paris $\rightarrow CaSO_4 \cdot \frac{1}{2}H_2O$. It is used in construction, casting, and medical applications. **Hence, pair 1 is correctly matched.**
 - Baking Soda $\rightarrow$ **Sodium bicarbonate ($NaHCO_3$)**, commonly known as baking soda or bicarbonate of soda, or salaratus, is a chemical compound with the formula $NaHCO_3$. It is a salt composed of a sodium cation and a bicarbonate anion. Thus, $NaHCO_3$ not Na_2CO_3 . Na_2CO_3 is Sodium carbonate (Washing soda), not Baking soda. **Hence, pair 2 is not correctly matched.**

- Washing Soda $\rightarrow \text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$. It is used in laundry detergents and water softening. **Hence, pair 3 is correctly matched.**
- Gypsum $\rightarrow \text{CaSO}_4 \cdot 2\text{H}_2\text{O}$. It is a naturally occurring mineral, used in cement, fertilizer, and drywall manufacturing. **Hence, pair 4 is correctly matched.**
- **Hence, option (c) is the correct answer.**

Q 64. B • Asexual reproduction is a mode of reproduction where offspring are produced by a single parent without the involvement of gamete formation and fusion. Offspring produced are genetically identical to the parent (clones) because there is no mixing of genetic material. **Hence, statement 1 is correct.**

- In contrast, sexual reproduction involves the formation of haploid gametes (sperm and egg cells) through meiosis, and their subsequent fusion (fertilisation) to form a diploid zygote, leading to genetic variation in the offspring. **Hence, statement 2 is correct.**
- While sexual reproduction is common in many multicellular organisms, many can also reproduce asexually, and some exclusively use asexual methods. Examples include many plants (vegetative propagation), fungi (spore formation), and some animals like starfish (fragmentation) or certain lizards (parthenogenesis). **Hence, statement 3 is not correct.**

Q 65. C • **Landsat Next** is a pioneering satellite program that will **map and observe the Earth's changing surface to support mining exploration, monitor the Earth's climate, water and environment, crop and agricultural health, and manage floods, fires, and natural disasters.** Landsat Next continues Landsat's decades-long data record of multispectral imagery, which affords global, synoptic, and repetitive coverage of Earth's land surfaces at a scale where natural and human-induced changes can be detected, differentiated, characterized, and monitored over time. **Hence statement 1 is correct.**

- **LandSat Next will be led by NASA and the US Geological Survey and joined by the Australian Government in 2024. Hence statement 2 is not correct.**
- **The launch of the Landsat Next mission is planned for late 2030/early 2031. Three identical satellites** are expected to make up Landsat Next, each carrying instruments that detect 26 **spectral bands spanning visible and infrared light.** This is more than double the spectral bands of the current Landsat 9 program. New bands will help scientists, farmers, and businesses, among others, to better observe, predict, and manage everything from algal blooms to crop stress, from snow and ice cover changes to bushfires. **Landsat Next will also gather higher-resolution and more detailed images of Earth's surface than ever. Hence statement 3 is correct.**
- **Hence option (c) is the correct answer.**

Q 66. B • Insulin is a peptide hormone produced by beta cells of the pancreatic islets encoded in humans by the insulin gene. It is the main anabolic hormone of the body.

- Insulin is a crucial hormone that regulates blood sugar (glucose) by helping move it from the bloodstream into cells for energy or storage, acting as the body's main fuel manager.
- In diabetes, the body either doesn't make enough insulin or can't use it properly, causing high blood sugar.
- Manufactured insulin, given via injection or pump, replaces or supplements the body's supply to control blood sugar, with different types varying in how fast they work.

- Hence, option (b) is the correct answer.

Q 67. A • Passive immunity is immediate but temporary protection from diseases, acquired by receiving pre-made antibodies from an external source, rather than producing them yourself, like a baby getting antibodies from its mother (placenta/breast milk) or someone getting an antibody injection (immune globulin) for immediate defense against toxins or viruses, offering quick relief but fading in weeks or months. **Hence, statement 1 is correct.**

- Passive immunity provides immediate protection but does not result in long-term immune memory. Since the recipient's immune system is not activated to produce its own antibodies, no memory cells are formed, and the protection is short-lived, lasting only a few weeks to months. **Hence, statement 2 is not correct.**

Q 68. C • A **transverse wave** is a wave that **vibrates perpendicular to the direction of the wave or path of propagation**. A simple example is given by the waves that can be created on a horizontal length of the string by anchoring one end and moving the other end up and down.

- In a transverse wave, the particles are displaced perpendicular to the direction the wave travels. In a longitudinal wave, the particles are displaced parallel to the direction the wave travels. An example of longitudinal waves is compressions moving along a slinky. **Hence statement 1 is not correct.**
- **Mechanical transverse waves cannot propagate in a gaseous or a liquid medium** because there is no mechanism for driving motion perpendicular to the propagation of the wave. **Hence statement 2 is correct.**
- On the other hand, the common electromagnetic waves (unlike mechanical waves), like visible light do not require a medium to travel and thus can travel in the gaseous medium as well as in vacuum.
- Transverse waves **commonly occur in elastic solids**; the oscillations, in this case, are the displacement of the solid particles away from their relaxed position, in directions perpendicular to the propagation of the wave. Since those displacements correspond to local shear deformation of the material, a transverse wave of this nature is called a shear wave. **In seismology, shear waves are also called secondary waves or S-waves. Hence statement 3 is correct.**

Q 69. A • **Pauli Engine, a quantum heat engine rooted in the Pauli exclusion principle.**

- Bosons are particles that can occupy the same quantum state, leading to phenomena like superfluidity and superconductivity. On the other hand, fermions, including electrons, follow the Pauli exclusion principle, preventing two identical fermions from occupying the same quantum state simultaneously.
- The Boson-Fermion Quantum Engine leverages the complementary characteristics of these particles. This innovative approach involves creating a system where bosons and fermions coexist and interact, leading to emergent behaviors that can be harnessed for various applications. **Emerging from this exploration is the intriguing concept of the boson-fermion quantum engine, a device that harnesses the unique properties of both bosons and fermions to perform energy conversion.**
- **Physicists in Germany have come up with a way to convert the energy difference between two quantum states of a group of atoms into work.** The device adapts the principles of the familiar classical engine to the subatomic realm, giving physicists a way to study the nascent field

of quantum thermodynamics in more detail as well as, possibly, build better quantum computers.

Hence, option (a) is the correct answer.

- Q 70. A** • The gravitational force between two bodies is directly proportionate to the product of masses of the two objects and inversely proportionate to the square of the distance between them.
- The electromagnetic force between two bodies is directly proportionate to the product of charges of the two objects and inversely proportionate to the square of the distance between them. **Hence statement 3 is not correct.**
 - Gravitational force is always attractive while Electromagnetic force can be both attractive and repulsive. The same charges repel each other while opposite charges attract. **Hence statement 1 is correct**
 - The electromagnetic force is not dependent upon mass. **Hence statement 2 is not correct.**
- Q 71. A** • **AMoRE** searches for the signature of **neutrinoless double beta decay** from the ^{100}Mo with an experiment on the scale of 100 kg of the isotope ^{100}Mo . The ultimate goal of the experiment is to achieve a sensitivity that covers the entire $\langle m\beta\beta \rangle$ range that is allowed by the inverted neutrino mass hierarchy. **Hence statement 1 is correct.**
- The AMoRE project is a series of experiments. The first phase of the AMoRE, called AMoRE-Pilot, has been operating since 2015 with about 1.5 kg of $^{40}\text{Ca}^{100}\text{MoO}_4$ crystals at **YangYang underground Laboratory (Y2L) in South Korea**. **Hence statement 2 is not correct.**
 - **Hence option (a) is the correct answer.**
- Q 72. A** • Filariasis refers to a group of parasitic infections caused by thread-like nematodes transmitted to humans through insect vectors such as mosquitoes, blackflies, and deerflies. These parasites are common in tropical and subtropical areas and cause diseases that are often preventable. Infections can range from having no symptoms at all to causing severe and visible physical damage that can lead to stigma and disability. **Hence, Pair 1 is not correctly matched.**
- Dengue is a viral infection caused by the dengue virus (DENV), which is transmitted to humans through the bite of infected mosquitoes. About half of the world's population is now at risk of dengue, with an estimated 100–400 million infections occurring each year. Dengue is found in tropical and sub-tropical climates worldwide, mostly in urban and semi-urban areas. **Hence, Pair 2 is not correctly matched.**
 - Chickenpox is a highly contagious disease caused by the varicella-zoster virus (VZV), resulting in an itchy, blister-like rash, fever, and fatigue, usually lasting 4–7 days. It spreads easily via airborne droplets (coughing/sneezing) or contact with blister fluid. Contagious 1–2 days before the rash appears until all blisters have scabbed, it is preventable through vaccination. **Hence, Pair 3 is correctly matched.**
 - Malaria is a life-threatening disease spread to humans by some types of mosquitoes. It is mostly found in tropical countries. It is preventable and curable. The infection is caused by a parasite and does not spread from person to person. Symptoms can be mild or life-threatening. Mild symptoms are fever, chills and headache. Severe symptoms include fatigue, confusion, seizures, and difficulty breathing. **Hence, Pair 4 is not correctly matched.**

- Q 73. A** • **Electric Fuse** is a current interrupting device that protects the electrical circuit.

- The principle behind the operation of the fuse is the **heating effect of the electrical current**. If the current passes through a conductor with a certain resistance, the loss due to the conductor's resistance is dissipated in the form of heat. Under normal operating conditions, the heat produced in the fuse element is easily dissipated into the environment due to the flow of current through it. As a result, the fuse part remains at a temperature below its melting point. **Hence statement 1 is correct.**
- Whenever some faults, such as short circuits, occur, the current flow through the fuse element exceeds the prescribed limits. This creates an excess of heat that melts the fuse part and breaks the circuit.
- So if the electrical current passing through the fuse goes beyond what the device can handle, the fuse link is melted and the circuit is opened to prevent damage to the electrical component.
- Fuses are always connected in series with the component to be protected from overcurrent, so that when the fuse blows (opens) it will open the entire circuit and stop current through the component. **Hence statement 2 is not correct.**

- Q 74. B**
- Square windows were hazardous for airplanes because the pressures inside and outside the cabin are so different, that the sharp edgy corners would not be able to handle the pressure difference and give in.
 - Sharp corners naturally concentrate more stress, which further gets weakened by the immense air pressure. Curved or round windows distribute the stress of the outside pressure of air, and thus are stronger and prone to deformation because of the pressure.
 - **Hence option (b) is the correct answer.**

- Q 75. A**
- **Electrolysis is a chemical process where an electric current is passed through an electrolyte (a conducting liquid) to drive a non-spontaneous reaction. It is used in various industrial applications, including electroplating, metal purification, and hydrogen production.**

Electrolysis involves passing an electric current through an ionic solution or molten compound to induce a chemical reaction.

- This is a widely used industrial process for extracting and refining metals, producing gases, and manufacturing chemical compounds. **Hence, statement 1 is correct.**

In the electrolysis of water:

- Hydrogen (H_2) is released at the cathode (-ve electrode).
- Oxygen (O_2) is released at the anode (+ve electrode). **Hence, statement 2 is not correct.**

Electroplating is a process where a thin layer of metal is deposited onto another metal surface using electrolysis.

- Metal purification (such as refining copper) also relies on electrolysis to remove impurities. **Hence, statement 3 is correct.**

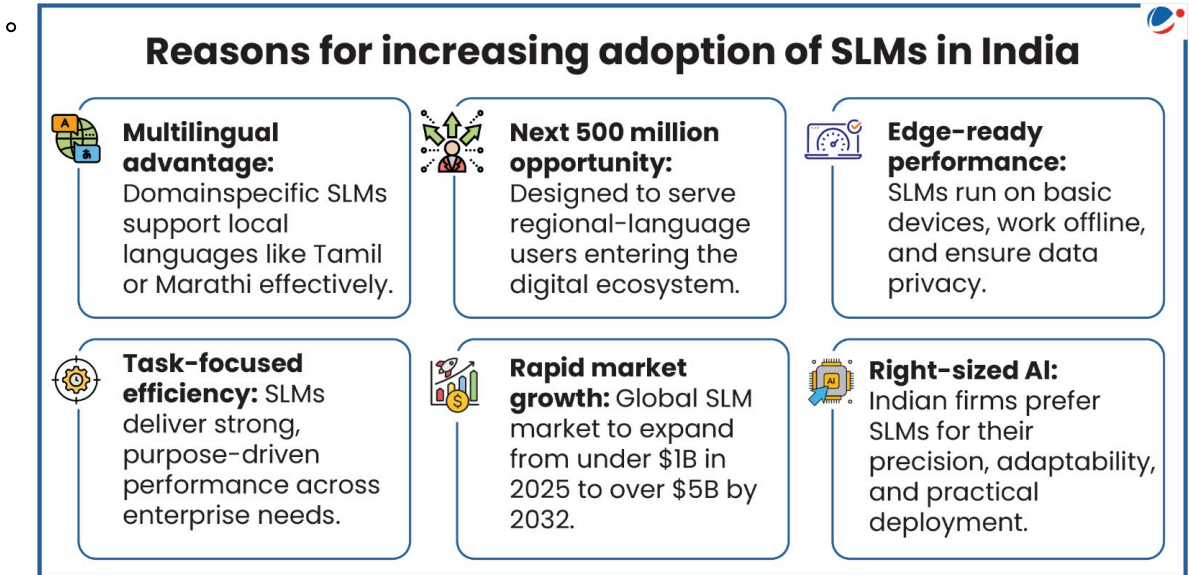
Green Hydrogen is produced by splitting water via electrolysis using renewable energy sources (solar, wind, hydro).

- Unlike gray hydrogen (produced from fossil fuels), Green Hydrogen is an eco-friendly alternative for energy storage and fuel.

- Q 76. A**
- **Recent Context:** The Union Minister for Electronics and IT stated that the future of AI will be shaped by SLMs rather than extremely Large Language Models (LLMs).

What are SLMs?

- SLMs are compact AI systems built on simpler neural network architectures, designed to generate and understand natural language, as LLMs do.
- Parameters used by SLMs: several million to 30 billion parameters, whereas LLMs often possess hundreds of billions or even trillions parameters.
- **At present, nearly 95% of AI work globally is currently handled by SLMs. E.g. Llama, Mistral, Gemma and Granite etc. Hence, option (a) is the correct answer.**



- Q 77. A** • A **hypersonic weapon** refers to one that travels faster than **Mach 5, or five times the speed of sound (330 m/s)**. There are two types: **Hypersonic glide vehicles (HGV)** are launched from a rocket, similar to regular ballistic missiles, before gliding to a target, whereas **Hypersonic cruise missiles** are powered throughout their flight via air-breathing engines called **Scramjets**, after acquiring their target. **Hence statement 1 is not correct.**
- Hypersonic weapons have been in existence since the middle of the 20th century. Ballistic missiles, for instance, have long possessed these speeds. What differentiates today's hypersonic capabilities is that, unlike ballistic missiles, HGVs do not follow a **parabolic trajectory** to their target, making their re-entry into the Earth's atmosphere much quicker and at a much lower altitude. They can subsequently glide to their target while executing advanced evasive maneuvers under guided flight. Additionally, HGVs can have ranges up to thousands of kilometers, effectively making them equivalent to **intercontinental ballistic missiles (ICBMs)**. Hypersonic cruise missiles fly at considerably lower altitudes, typically around 200 feet above the ground, and also possess enhanced maneuverability and speed. Consequently, hypersonic weapons can bypass most traditional **missile defense systems**, and their detection poses a serious challenge. For example, **terrestrial-based radar** cannot detect them until late in their flight. The threat is only bolstered by the fact that they are capable of carrying both traditional and nuclear warheads. **Hence statement 2 is correct.**
 - **Avangard is a hypersonic glide vehicle developed by Russia.** Avangard can presumably reach speeds up to Mach 20 and can be used to deliver nuclear and conventional payloads. **Hence statement 3 is not correct.**

- Q 78. B** • The cell wall is a rigid structure in plant cells (also fungi, bacteria, and some protists) that provides mechanical strength, structural support, and protection against mechanical damage, infection, and osmotic stress. **Hence, statement 1 is correct.**

- The cell wall is a non-living, rigid structure located outside the living plasma (cell) membrane. **Hence, statement 2 is correct.**
- The cell wall is a fully permeable or freely permeable structure, meaning it allows most substances, including water, minerals, and other small molecules, to pass through freely. It does not control selective permeability; this function belongs to the living plasma membrane which lies inside the cell wall. **Hence, statement 3 is not correct.**

Q 79. B • Nakhchivan is a landlocked exclave of the Republic of Azerbaijan. It is physically separated from the rest of Azerbaijan by a strip of Armenian territory (the Syunik Province). **Hence, option (b) is the correct answer.**

- It is located in the South Caucasus region, at the crossroads of Eastern Europe and Western Asia.
- Nakhchivan often appears in the news in the context of Azerbaijan-Armenia relations, regional connectivity projects, and Caucasus geopolitics.



Q 80. B • The Nancy Grace Roman Space Telescope, also known as "Roman" or the "Roman Space Telescope," is NASA's future infrared space observatory that will attempt to tackle some of the most pressing questions in cosmology by addressing the **mysteries of dark energy** and will also search for planets outside the solar system — **exoplanets**, and will investigate the **physics of distant stars**. **Hence statement 1 is not correct.**

- **Set to launch around 2026**, the ROMAN space telescope possesses a wide field of view, allowing it to generate never-before-seen big pictures of the universe essential for tackling some of the most pressing cosmic mysteries. The Roman space telescope will be at **Lagrange point 2**, a stable gravitational point between Earth and the sun located around 1 million miles (1.5 million kilometers) from our planet. **Hence statement 2 is correct.**
- The Nancy Grace Roman Space Telescope began life as the **Wide Field Infrared Survey Telescope (WFIRST)** in **2010**, only gaining its current name a decade later when, in May 2020, it was

renamed in honor of **Nancy Grace Roman**, a pioneering scientist who served as NASA's first chief astronomer from 1961 to 1963. During her life, Roman was affectionately known as "**the mother of Hubble**." The nickname emerged due to the fact that Roman tirelessly advocated for new tools that would allow scientists to study the broader universe, which led to the launch of the Hubble Space Telescope in 1990. **Hence statement 3 is correct.**

- **Hence option (b) is the correct answer.**

Q 81. B National Manufacturing Policy:

- The Government has set a growth target for the manufacturing sector from 16% of GDP to reach 25% of GDP by 2025. The National Manufacturing Policy and the emphasis on "Make in India" is a powerful strategy for economic development. In accordance with this objective, DST has initiated this program for developing Advanced Manufacturing Technologies in October 2015.

Objectives

- Five thrust areas were selected for developing novel manufacturing technologies :
- Nano materials & surfaces,
- Robotics & automation
- Precision manufacturing,
- Manufacturing process of Pharmaceuticals & Bio manufacturing
- Advanced forming & near net shaped processing.

Near net shaped processing:

- Near netshape is an innovative concept in industrial manufacturing. The main focus of this technology is to produce parts, as near as possible close to their final shape and contour, implementing non-chipping techniques. In this way the manufacturing gives the possibility of a finished product with minimal cutting. Near netshape technology also generates the opportunity to reduce the productive steps for a given process chain. Both the above mentioned characteristics have the same goal: achieving cost reduction. This will lead to quality improvement of the finished product.
- **Hence, option (b) is the correct answer.**

- Q 82. C**
- The electrostatic forces in ionic compounds are quite strong. While the covalent bonds within molecules are strong, the forces between the separate molecules are weak, leading to lower melting/boiling points. In ionic compounds, the molecules consist of oppositely charged ions held together by very strong electrostatic forces in a rigid crystal lattice, requiring a large amount of energy to overcome these powerful attractions and break the structure.
 - Bonds formed by sharing an electron pair between two atoms are covalent bonds. **Covalently bonded molecules are seen to have strong bonds within the molecule, but intermolecular forces are weak.** This gives rise to **low melting and boiling points** of these compounds. **Hence, the assertion is correct, and the reason is incorrect.**
 - **Hence option (c) is the correct option.**

- Q 83. C**
- While it is stressful to watch TV in darkness owing to the constant relaxation and contraction of the iris and hence recommended not to watch TV in darkness, the same does not apply to watch a movie in a dark theatre.

- The simple reason is that movie, unlike the TV or computer image is not in the form of pixels. The movie (**in a traditional theatre**) is in the form of analog. **Hence statement 1 is correct.**
- in the case of movies, the stress caused to the eyes is less as the eyes' field of view is completely occupied by the screen. **Hence statement 2 is correct.**
- This is not the case with a TV or computer. The screen is smaller and does not fill the complete field of vision of the eyes.
- Moreover, the object on the TV is constantly flickering and we see areas that are of different light intensities.

Q 84. D • Recent Context: 10 Years of Startup India Initiative

About Startup India Initiative

- It is flagship initiative aimed at supporting entrepreneurs, building a robust startup ecosystem and transforming India into a country of job creators instead of job seekers.
- Launched: 16th January, 2016 (celebrated as National Startup Day)
- Nodal Agency: Department for Promotion of Industry and Internal Trade (DPIIT), Ministry of Commerce & Industry.

Decadal Achievements (2016–2026)

India is the world's 3rd-largest startup ecosystem.

- ~2.09 lakh DPIIT-recognised startups (2016-25)
- 120+Unicorns with a combined valuation exceeding \$350 Billion.

Employment Generation and Inclusivity: Over 21 Lakh jobs created, with a significant multiplier effect.

- ~45% of recognized startups have at least one-woman director/partner.
- ~50% of startups are based in Tier-II/III cities.

Other Schemes promoting Startup Ecosystem

- **Atal Innovation Mission (AIM),**
- **GENESIS (Gen-Next Support for Innovative Startups),**
- **NIDHI (National Initiative for Developing and Harnessing Innovations),**
- **ASPIRE (A Scheme for Promotion of Innovation, Rural Industries and Entrepreneurship). Hence, option (d) is the correct answer.**

- Q 85. A •** If an object of mass, m is falling freely from a height, h in a vacuum. **At the start, the potential energy is mgh and kinetic energy is zero.** It is zero because its velocity is zero. The total energy of the object is thus mgh . As it falls, its potential energy will change into kinetic energy. If v is the velocity of the object at a given instant, the kinetic energy would be $\frac{1}{2}mv^2$. **Hence option (a) is correct.**
- **As the fall of the object continues, the potential energy would decrease while the kinetic energy would increase.** When the object is about to reach the ground, $h = 0$ and v will be the highest. Therefore, the kinetic energy would be the largest and potential energy the least. However, **the sum of the potential energy and kinetic energy of the object would be the same at all points.** That is, potential energy + kinetic energy = constant, or $mgh + (1/2)mv^2 = \text{constant}$

- The sum of kinetic energy and potential energy of an object is its total mechanical energy. During the free fall of the object, the decrease in potential energy, at any point in its path, appears as an equal amount of increase in kinetic energy. (Here the effect of air resistance on the motion of the object is zero as the object is in a vacuum). There is thus a continual transformation of gravitational potential energy into kinetic energy.
- Whenever energy gets transformed, the total energy remains unchanged. This is the law of conservation of energy. According to this law, energy can only be converted from one form to another; it can neither be created nor destroyed. The total energy before and after the transformation remains the same. The law of conservation of energy is valid in all situations and for all kinds of transformations.

Q 86. C • GPS is a satellite-based radio navigation system that provides accurate location and time information worldwide. Hence, statement 1 is correct.

GPS is owned and operated by the United States Department of Defense (DoD).

- The initial development of GPS began in the 1970s under the codename Navstar GPS.
- The system became fully operational in 1995 with a constellation of 24 satellites in orbit.
- GPS works by utilizing a network of satellites that transmit precise timing signals to receivers on Earth.

It works on the principle of trilateration. It uses the signals from multiple satellites to determine the location of a receiver on Earth. Hence, statement 2 is correct.

- Knowing its distance to 3 or more satellites and their positions in orbit, the receiver can generate three spheres with the satellites as their centers and the measured distances as their radii.
- The point where these spheres intersect is the approximate location of the receiver.
- The receiver also uses the satellite signals to obtain a precise time reference, enabling it to calculate its elevation (height above sea level)

GPS is a network of 24 satellites.

- These satellites are arranged in six different orbital planes, with four satellites in each plane.
- The satellites are arranged in six equally-spaced orbital planes, with four baseline satellites in each plane.
- This 24-slot arrangement ensures users can view at least four satellites from virtually any point on the planet. **Hence, statement 3 is correct.**

Q 87. A • Aqua regia, (Latin for 'royal water') is a freshly prepared mixture of concentrated hydrochloric acid and concentrated nitric acid in the ratio of 3:1. Hence statement 1 is not correct.

- It can dissolve gold, even though neither of these acids can do so alone. Aqua regia is a highly corrosive, fuming liquid. It is one of the few reagents that is able to dissolve gold and platinum. **Hence statement 2 is correct.**

Q 88. A • Recent Context: Venezuelan Opposition Leader agreed to offer her Nobel Peace Prize Medal to U.S. President

About Nobel Peace Prize

- Awarded to person "who shall have done the most or the best work for fraternity between nations, for the abolition/reduction of standing armies and for holding and promotion of peace

congresses" (As per the Will of Alfred Nobel).

- **Awarded by: Norwegian Nobel Committee since 1901. Hence, statement 1 is correct.**

Criteria: Since World War II, it is awarded to honour efforts in four main areas:

- Arms control and disarmament, peace negotiation, democracy and human rights, and work towards creating a better organized and more peaceful world.
- In the 21st century, efforts to limit harm done by man-made climate change and threats to environment are too considered relevant to the prize.

Inclusions of the Prize: Medal, a diploma, and a cheque (currently 11 million Swedish crowns).

- The medal is currently, 18 carats gold, 196 grams with diameter of 6.6 centimetres.
- Awardees from India: Mother Teresa (1979); 14th Dalai Lama (1989); Kailash Satyarthi (2014).

Transferability of the Prize and the Medal

- **Finality of the Prize: Once announced, it cannot be revoked, shared, or transferred with the decision being final and standing for all time.**

Transferability of the Medal: While the honour and title are non-transferable, the medal can be gifted, loaned or displayed by someone else. Hence, statement 2 is not correct.

- E.g., Dmitry Muratov's medal was auctioned for over USD 100 million to support refugees from Ukrainian War.

Q 89. D Absciscic acid (ABA)

- It is widely recognized as the "stress hormone" in plants because its synthesis is rapidly stimulated by environmental stressors like drought, high salinity, and cold temperatures.

It helps plants survive these adverse conditions by triggering several adaptive responses:

- Stomatal Closure: ABA signals the guard cells to close the stomata, which significantly reduces water loss through transpiration.
- Dormancy Induction: It promotes seed and bud dormancy, ensuring that they only germinate or sprout when conditions are favorable for survival.
- Growth Inhibition: ABA acts as a general growth inhibitor, helping the plant conserve energy and resources when conditions are not ideal for active growth.

Hence, option (d) is the correct answer.

- Q 90. D • **Statement 1 is not correct:** A water molecule contains two Hydrogen atoms and an Oxygen atom. Hydrogen has three isotopes; protium (99.98%), deuterium and tritium. Protium has one electron and one proton. Deuterium has a neutron in the nucleus in addition to the electron and the proton. Deuterium is as twice as heavier than the most abundant Hydrogen atom. **Heavy water contains a large proportion of deuterium atoms than the usual Hydrogen atom.** Therefore, its molecular weight and the density is higher than the normal water.

- The hard water contains more minerals, especially Magnesium (Mg) and Calcium (Ca) than the normal water.
- **Statement 2 is not correct: Heavy water is not radioactive.** It is an oxide of deuterium (D₂O) which is a stable isotope of hydrogen. The hardness of the hard water is greater than the hardness of the normal water. Hard water does not cause any harmful effect on human health, but it causes so

many additional problems such as leaving white coloured deposits in cooking or boiling equipment, bathroom floors and in water pipes.

- **Statement 3 is not correct: Moderator is required in a thermal reactor to slow down the neutrons produced in the fission reaction to 0.025 eV (thermal reaction) so that the chain reaction can be sustained.** A good moderator should have excellent slowing down power and low absorption cross-section for neutrons. Different moderators normally in use are Heavy Water, Graphite, Beryllium and Light water. Heavy Water is an excellent moderator and is used in Pressurized heavy-water reactor. Water quality has a major effect on the integrity of piping in boiling water reactors (BWRs) and steam generators in pressurized water reactors (PWRs). **Hard water is not used in a nuclear power plant.** The reason is that any mineral impurities will become very radioactive and contaminate plant systems making maintenance more expensive. Also, in BWRs, the mineral impurities would concentrate on the reactor vessel and create huge (thousands of pounds) mineral deposits. The purest water possible, i.e. deionized reactor grade water, is used in PWRs and BWRs.
- Tritium the third isotope of Hydrogen which is very rare is radioactive (it has two neutrons and one proton).

- Q 91. A**
- The skin forms the body's primary physical barrier, a tough outer layer that prevents most pathogens from entering the body. **Hence, Pair 1 is not correctly matched.**
 - Saliva, tears, and other bodily secretions are considered physiological (or chemical) barriers. They contain enzymes like lysozyme, which chemically break down the cell walls of bacteria, thus inhibiting microbial growth.
 - Macrophages are a type of white blood cell that performs phagocytosis (engulfing and digesting pathogens and cellular debris). They are a key component of the cellular innate immune response. **Hence, Pair 2 is correctly matched.**
 - Chemical mediators of innate immunity are endogenous and exogenous molecules, including proteins, peptides, and fatty acids, that provide immediate, nonspecific defence against pathogens. **Hence, Pair 3 is not correctly matched.**
 - **Antibodies (immunoglobulins) are large proteins produced by B cells in the adaptive immune system.**

- Q 92. C**
- **Option 1 is not correct: Melting of ice is a physical change** because it involves change only in the physical state of water. Ice in solid state convert to water in the liquid state. Moreover, no new chemical substance is formed in the process and this change is reversible.
 - **Option 2 is correct:** Vinegar is an aqueous solution of acetic acid and trace chemicals that may include flavourings. Vinegar typically contains 5–20% by volume acetic acid. Usually, the acetic acid is produced by the fermentation of ethanol or sugars by acetic acid bacteria. **Fermentation is the chemical breakdown** of a substance by bacteria, yeasts, or other microorganisms.
 - **Option 3 is not correct:** If the chemical formulas do not change, then any changes are physical changes. **An example of physical is dissolving sugar in water.** Once the sugar has all dissolved, you have molecules of sugar mixed in with the molecules of water. The chemical formulas at the end are the same as they were at the start, so this is a physical change.

- Q 93. D** • Google announced a significant advancement in quantum computing as it unveiled its next-generation chip, 'Willow.' This new chip developed at the company's quantum lab in Santa Barbara, California, solved a complex mathematical problem in just five minutes — a task that would take classical computers longer than the universe's history.
- Quantum computing is a field of computer science that uses quantum mechanics to address problems that classical computers cannot solve. **Unlike classical computers, which use bits to represent information as 0 or 1, quantum computers rely on quantum bits or qubits.**
 - Hence option (d) is the correct answer.**
- Q 94. D** • Pepsin is the only enzyme that is secreted in an inactive form (as pepsinogen by chief cells of the gastric glands) and is activated in the stomach's highly acidic environment by hydrochloric acid (HCl). **Hence option 1 is correct.**
- Trypsin is a protein-digesting enzyme, but it is produced by the pancreas and released into the small intestine in its inactive form, trypsinogen. It is activated in the small intestine by an enzyme called enteropeptidase, and it functions optimally in the alkaline environment there (pH 7.5-8). **Hence option 2 is not correct.**
 - Rennin (also known as chymosin) is a milk-digesting enzyme found in the gastric juice of infants and helps in the coagulation and digestion of milk proteins. Like pepsinogen, its precursor (prorenin) is activated by stomach acid. It is generally not considered a significant enzyme in protein digestion in adult humans. **Hence option 3 is correct.**
- Q 95. D** **The tokamak is a promising technology for harnessing the power of nuclear fusion to generate clean and sustainable energy.**
- Fusion is a process where atomic nuclei combine to form heavier nuclei, releasing enormous amounts of energy.
- EAST (Experimental Advanced Superconducting Tokamak) is a tokamak fusion reactor located in China.**
- It is one of the most advanced tokamak reactors in the world and has achieved significant milestones in fusion research.
 - It uses a magnetic field to confine a plasma inside a doughnut-shaped vacuum vessel. **Pair 1 is correctly matched.**
- European Union: JET (Joint European Torus)**
- JET is currently undergoing a major upgrade that will increase its fusion power output by a factor of ten. **Pair 2 is correctly matched.**
- The JT-60 (Japan Torus-60) was a large tokamak fusion research facility located in Naka, Japan.**
- JT-60 was decommissioned in 2008 and replaced by a more advanced tokamak, JT-60SA, which is still under construction. **Pair 3 is correctly matched.**
- Tore Supra is an advanced tokamak (magnetic confinement fusion device) located at Cadarache, France. Pair 4 is correctly matched.**
- Q 96. A** • **The human genome is the entire set of deoxyribonucleic acid (DNA) residing in the nucleus of every cell of each human body. It carries the complete genetic information responsible for the**

development and functioning of the organism. The DNA consists of a double-stranded molecule built up by four bases – adenine (A), cytosine (C), guanine (G), and thymine (T). Every base on one strand pairs with a complementary base on the other strand (A with T and C with G) In all, the genome is made up of approximately 3.05 billion such base pairs.

- **While the sequence or order of base pairs is identical in all humans**, compared to that of a mouse or another species, there are differences in the genome of every human being that make them unique. The process of deciphering the order of base pairs, to decode the genetic fingerprint of a human is called genome sequencing.
- **Genome sequencing has been used to evaluate rare disorders**, preconditions for disorders, even cancer from the viewpoint of genetics, rather than as diseases of certain organs. Nearly 10,000 diseases — including cystic fibrosis and thalassemia — are known to be the result of a single gene malfunctioning.
- **IndiGen program aims to undertake whole genome sequencing** of thousands of individuals representing diverse ethnic groups from India. The objective is to enable genetic epidemiology and develop public health technology applications **using population genome data. Hence statement 1 is not correct.**
- **The Department of Biotechnology (DBT)** recently said that the exercise was to **sequence 10,000 Indian human genomes and create a database under the Centre-backed Genome India Project.** The Genome India project was approved by the government in 2020 with the aim of creating a comprehensive catalogue of **genetic variations found in the Indian population. A map of genetic diversity is essential for understanding the history of our evolution**, discovering the genetic basis for various diseases, and creating therapies of the future. **Hence statement 2 is not correct.**
- **The Human Genome Project (HGP)** is one of the greatest scientific feats in history. The project was a voyage of biological discovery led by an international group of researchers looking to comprehensively study all of the DNA (known as a genome) of a **select set of organisms. Launched in October 1990 and completed in April 2003, the Human Genome Project's signature accomplishment** – generating the first sequence of the human genome – provided fundamental information about the human blueprint, which has since accelerated the study of human biology and improved the practice of medicine. **Hence statement 3 is correct.**
- **Hence option (a) is the correct answer.**

- Q 97. D**
- The separation of visible light into its different colors is known as **Dispersion**. Each color is characteristic of a distinct wave frequency; and different frequencies of light waves will bend varying amounts upon passage through a prism. **Total Internal reflection** is complete reflection of a ray of light within a medium such as water or glass from the surrounding surfaces back into the medium. The phenomenon occurs if the angle of incidence is greater than a certain limiting angle, called the critical angle. **Refraction** is the change in direction of a wave passing from one medium to another caused by its change in speed.
 - A rainbow is a natural spectrum appearing in the sky after a rain shower. It is caused by **dispersion of sunlight by tiny water droplets**, present in the atmosphere. A rainbow is always formed in a direction opposite to that of the Sun. The water droplets act like small prisms. They **refract** and **disperse** the incident sunlight, then **reflect it internally**, and finally refract it again when it comes out of the raindrop. Due to the **dispersion of light and internal reflection**, different colours reach the observer's eye.

- Hence option (d) is the correct answer.

- Q 98. B**
- Enzymes act as biological catalysts and speed up biochemical reactions by lowering the activation energy required for the reaction to occur. **Hence, statement 1 is correct.**
 - Enzymes are not consumed during biochemical reactions. They are biological catalysts that remain unchanged after converting the substrate into products and are reusable for subsequent reactions. **Hence, statement 2 is not correct.**
 - Enzyme activity is significantly influenced by environmental factors such as temperature and pH. Each enzyme has an optimal temperature and pH, and deviations from these conditions can lead to decreased activity or denaturation. **Hence, statement 3 is correct.**

- Q 99. B**
- **Recent Context:** The Sukanya Samriddhi Yojana (SSY) has completed 11 years since its launch under the Beti Bachao, Beti Padhao campaign.

About Sukanya Samriddhi Yojana (SSY)

- **Launch:** 22nd January 2015 under Beti Bachao, Beti Padhao. **Hence, statement 1 is correct.**
- **Objective:** Promote welfare of the girl child and create a long-term savings corpus for education and future needs.

Key features

- **Eligibility:** Girl child below 10 years;
- **Only one SSY account is permitted per girl child, and a family can open accounts for a maximum of two girl children. Hence, statement 2 is correct.**
- **Annual deposit** of ₹250–₹1.5 lakh for 15 years.
- **Account operated** by guardian till 18 years.
- **Maturity & withdrawal:** Matures after 21 years; up to 50% withdrawal for education after 18 years is allowed. **Hence, statement 3 is not correct.**

- Q 100. A**
- **Space stations allow astronauts to conduct research and experiments in space over a long duration.** The largest and most sophisticated of these is the International Space Station (ISS). It is a habitable artificial satellite, in low Earth orbit (at an altitude of between 370–460 km). Key partners for ISS:

- European countries (represented by European Space Agency)
- United States (National Aeronautics and Space Administration)
- Japan (Japan Aerospace Exploration Agency)
- Canada (Canadian Space Agency) and
- Russia (Roscosmos)

- **Aside from the ISS, three nations (U.S., Russia and China) have independently launched and operated space stations.**
 - **Russia:** Salyut 1 was the world's first space station launched by the Soviet Union on April 19, 1971. Then Russia launched various space stations from Salyut 2 to Salyut 7, and Mir.
 - **China:** Tiangong 1 launched in 2011, Tiangong -2 launched in 2016 were test space labs and Tiangong space station was launched in 2021 (fully operational since late 2022). **Hence, option (a) is the correct answer.**

- **USA: Skylab:** USA's first space station, launched by NASA in 1973.
- **Future missions:**
 - **NASA's Artemis Base Camp** on the moon's surface and the Gateway in lunar orbit.
 - **India's Bharatiya Antariksha Station.**

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